From Social Navigation Fundamentals to Novel Humanoid Autonomy: An In-Depth Thematic and Systematic Evidence Review

Geoff-Sim research synthesis and novelty roadmap

Expanded edition: 5 October 2026
Literature-discovery cutoff: 22 September 2026

Abstract

Geoff-Sim integrates Unitree G1 locomotion, MuJoCo, ROS 2, online mapping, pedestrian tracking, human-aware planning, collision supervision, and experimental benchmarking. This review evaluates 50 identity-verified research publications and preprints spanning 1995–2026, with selected primary full-text inspection for 46 works and explicit record-limited appraisals for four. It connects classical collision avoidance, learning-based crowd navigation, trajectory prediction, legged control, onboard perception, and social-navigation evaluation. Recent work already demonstrates LiDAR-conditioned humanoid avoidance, cross-embodiment navigation, and physically simulated multi-task benchmarks. Consequently, the strongest positioning is not the first integration of walking and social navigation. It is a falsifiable investigation of how calibrated locomotion response, observation age, uncertainty, and planning/supervision interaction affect closed-loop collision risk and mission liveness on a walking humanoid. The repository has unusually useful information-source and experiment-validity instrumentation, but its documented onboard trials still include collisions and timeouts. Five research opportunities are ranked, with explicit baselines, ablations, metrics, and claim boundaries. Source identity verification is separated from abstract-level screening, selected full-text analysis, and experimental replication; the latter was not undertaken.

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How to read this single PDF. Start with the executive question and the fundamentals narrative in Section 3. Follow the method families and the deeper comparisons in Section 10; then use the ranked gaps and Section 13 to see what must be demonstrated for a defensible contribution. Appendix B gives all 50 structured critical appraisals. Citations link to the complete bibliography, publication records link out to DOI/arXiv, and bibliography entries link back to their citing pages. The table of contents and PDF bookmarks are navigable.

1 Executive assessment

Recommended core research question: *Can a calibrated, observation-age-aware model of a humanoid's executable motion improve the collision–completion trade-off of social navigation under partial observability, relative to matched geometric, temporal, and fixed-margin baselines?*

This question fits the current implementation closely. It makes the robot's actual delayed response, rather than its requested planar velocity, the object of prediction. It also recognizes that a planner using a fresh message can still be acting on an old observation. Its novelty must be established by the specific model, calibration procedure, decision rule, and controlled evidence, because dynamics-aware avoidance and biped crowd navigation are established topics [6, 15, 16].

Three conclusions organize the review:

1. **The field has moved beyond isolated obstacle avoidance.** Recent systems combine learned locomotion with perception and navigation; the closest humanoid paper maps LiDAR and proprioception directly to motor commands, and FLUX reports transfer to a physical G1 [18, 48]. A general claim that previous research ignores humanoids, dynamics, or raw sensing is untenable.
2. **The remaining problem is measurable closed-loop reliability.** Forecast accuracy, nominal command feasibility, or a visually successful demonstration does not establish low collision risk over complete missions. Perception loss, stale observations, tracking association, command shaping, and stopping transients can interact. Current repository documentation provides concrete examples of this gap.
3. **Evaluation can be a substantive contribution if it tests a new scientific question.** Separating oracle control, sensor-only control, and independent ground-truth scoring enables causal diagnosis. The contribution would be a validated protocol and new findings about failure mechanisms, not simply another simulator or another collection of ROS nodes.

The most defensible near-term paper is a simulation-based methods and evaluation study. Hardware transfer and human comfort should be treated as subsequent validation targets, not inferred from MuJoCo clearance measurements.

2 Scope, search procedure, and verification standard

2.1 Review type and research questions

The original discovery was a **structured critical scoping search**. This expanded edition applies **systematic extraction and thematic synthesis to that defined corpus**; it is not an exhaustive database review or a meta-analysis. It asks:

1. What is already established about planning around people while respecting the dynamics and sensing constraints of a mobile or legged robot?

2. Which recent humanoid and cross-embodiment systems are direct competitors?
3. What evidence distinguishes geometric avoidance, social acceptability, and reliable mission-level autonomy?
4. Which gaps can this repository realistically investigate, and what would falsify the proposed advantages?

The search cutoff is 22 September 2026. Foundational work is retained when it defines a necessary baseline or assumption. The latest included preprint, DPed-VLN, was submitted on 18 September 2026 [14]; the expansion inspects its primary text without treating a recent preprint as mature validation. Date and venue are taken from retrieved records, not inferred from identifier prefixes.

2.2 Search and inclusion procedure

Discovery used Crossref title/bibliographic searches, arXiv title and topic searches, and references encountered in closely related papers. OpenAlex was used as an auxiliary locator for an accessible copy of the older biped paper. Representative topic strings were "social humanoid navigation", "socially aware robot navigation", "safe navigation biped crowd", "LiDAR legged collision avoidance", "humanoid sim-to-real", and exact method titles. The reproducible query families and screening decisions are recorded in `SEARCH_LOG.md`. Exact selected identifiers are in `sources.json`.

Inclusion required a direct connection to at least one of: human-aware navigation; crowd perception/prediction; executable collision avoidance; legged/humanoid control; simulation/middleware used by the project; or evaluation methodology. Foundational infrastructure papers are included to separate enabling technology from methodological novelty. Generic language models, manipulation-only systems, and unrelated humanoid demonstrations were not used to inflate the count. Title-search mismatches and duplicate versions were rejected. For example, a search for FAST-LIO2 returned both an SSRN record and the IEEE article; the IEEE record was selected.

The retained corpus contains **50 distinct works: 41 Crossref DOI records and nine arXiv records**. An arXiv citation does not imply that the work is unpublished; it means the cited version was verified through arXiv rather than an independently checked proceedings record. Publication-year differences between preprints, early access, and journal issues are normal. This review uses the year returned for the selected record.

2.3 What "verified" means here

Verification has three distinct levels:

Identity verification (all 50). A primary bibliographic record was retrieved from Crossref or arXiv. Title, author list, year, DOI/arXiv identifier, and source URL were extracted and checked. The evidence snapshot records the retrieval timestamp and SHA-256 of the response. This establishes that a matching paper record exists; it does not establish the truth of its findings.

Content inspection. The expanded edition adds selected full-text inspection for 46 works, including foundations, surveys, methods, datasets and close competitors. Each appraisal identifies its inspected material. Four unavailable originals remain record-limited. Specific table-level observations refer to the inspected copies; a preprint is not silently equated with every later version of record.

Replication. No external algorithm was reimplemented or rerun for this review. No new robot experiment or human-subject evaluation was performed. Repository results are explicitly

attributed to existing development records.

`verified_sources.json` retains the identity evidence and available abstracts. `READING_AUDIT.md` preserves the original audit; `FULLTEXT_AUDIT.md` and `paper_appraisals.json` record the expanded inspection. `verify_sources.py` checks the identity snapshot, bibliography and citation keys offline. The 2016 biped manuscript was successfully retrieved in this expansion and now receives a detailed appraisal. Search ranking, indexing delays, language coverage and single-workflow synthesis still limit completeness. Publication work should extend citation screening and independently review the novelty claims.

2.4 Expanded systematic extraction within the defined corpus

The expansion performed on 5 October 2026 changes the depth of inspection, not the declared September literature-discovery cutoff. The unit of inclusion remains the 50 distinct works in the original identifier manifest. All 50 receive the same structured appraisal fields; 46 have selected primary full-text passages inspected, while four retain explicitly limited evidence. A source being available as a PDF is not counted as an independently replicated result or as proof that every appendix has been read.

This is a **systematic evidence review of a defined, purposively assembled corpus**, with thematic synthesis. It is not an exhaustive systematic search of the entire robotics literature. The original discovery process did not retain a complete export of all database hits, and there was no independent dual screening. Those facts cannot be repaired by retrospectively inventing a PRISMA count or calling an automated download a second reviewer. The systematic contribution of this expansion is the explicit extraction instrument, source-level access record, consistent critical questions and traceable synthesis across every retained work.

Table 1: Corpus accounting for this expansion. These are review-workflow counts, not counts of every potentially relevant paper in the field.

Stage	Number
Inherited, identifier-verified unique works	50
Works retained for structured appraisal	50
Corresponding full texts retrieved and selected passages inspected	46
Original papers remaining record-limited	4
Independent external algorithm replications performed	0
New human-subject experiments performed for this review	0

2.5 Retrieval, version control and access limitations

Existing arXiv identifiers and corresponding-preprint locators were combined with DOI-matched OpenAlex location records, institutional repositories and author copies. OpenAlex was used to locate documents, not as the source of experimental results. Accessible PDFs were cached locally, converted to text and inspected at the loci recorded in `paper_appraisals.json`. URLs, timestamps, failed attempts and content hashes are retained in `fulltext_retrieval.json`. The copyrighted primary PDFs remain in the ignored local build cache; the delivered PDF contains original analysis and links, not a redistributed bundle of the papers.

An author manuscript and a publisher version can differ. For example, the retrieved DS-RNN text identifies a later arXiv revision than the conference year, while the Parkour manuscript predates its journal publication. Social Force also has a later repository deposit than its original journal date. The bibliography continues to identify the selected work; the appraisal identifies the inspected copy. Fine-grained claims belong to that copy until checked against the final

version. Unversioned repository URLs are additionally fingerprinted by their retrieved PDF hash.

The original full texts that remain unavailable in this pass are MuJoCo (2012), the 2013 sparse timed-trajectory paper, the 1998 velocity-obstacle paper and the 2013 dense-crowd cooperation paper. Their identities are verified. Available abstracts, institutional records and author software descriptions support bounded context, but they are not silently substituted for original methods and results. In particular, no new table-level quantitative claim is extracted from those unavailable texts. The four limitations are visible in the appraisals and audit.

2.6 Extraction instrument and thematic coding

Every paper is appraised using the following fields:

1. **Problem and contribution:** what question it addresses and which part of the autonomy pipeline it changes.
2. **Mechanism:** representation, policy/planner structure, information inputs, action interface and important modeling assumptions.
3. **Evidence:** the inspected study type, scenario, comparator, measurement boundary and reported result where directly supported.
4. **Critical appraisal:** limitations in internal validity, comparability, deployment relevance or interpretation; these are identified as the review's analysis rather than attributed wholesale to the authors.
5. **Novelty implication:** what the paper rules out as new, and which more specific experiment or contribution it motivates.
6. **Source anchor:** inspected sections/tables, source copy and reading level, so the claim can be traced back to material rather than a citation count.

The appraisals are grouped into eight analytical themes: social requirements; geometric interaction and safety; learned interaction/prediction; perception and estimation; datasets/evaluation; locomotion/transfer; sensor-level humanoid navigation; and enabling systems. These analytical themes overlap in substance: the grouping is a reading order, not a claim that each method belongs to only one scientific category. The separate web resource index uses a different set of navigation-oriented clusters for discoverability.

The synthesis asks the same cross-cutting questions across themes. Is an agent state observed, inferred or privileged? Does the action mean velocity, trajectory, joint target or torque? Is the crowd reciprocal, replayed or independent? Does "collision" mean contact, body overlap or a distance intrusion? Does runtime measure inference, an entire callback or the age of the observation at actuation? Are real people providing outcome judgments, or are geometric proxies being used? These distinctions make apparently similar success numbers scientifically nonexchangeable.

2.7 Quality appraisal without a misleading total score

The review does not assign a synthetic "quality score" to dissimilar papers. A theorem, a dataset, a taxonomy and a hardware deployment supply different forms of evidence. Instead, six appraisal dimensions are applied where relevant:

Source validity. Are sensing inputs, privileged training inputs and evaluation truth identifiable and separated?

Comparator validity. Do baselines share task conditions, tuning budget, observations and unavoidable actuation limits? Are adapters disclosed?

Execution validity. Is the claimed motion physically executed, and are its dynamics and contact model appropriate to the claim?

Temporal validity. Are acquisition time, processing delay and applied command identity distinguishable?

Statistical validity. Are exposure, denominators, repeats, uncertainty and failed/invalid trials described sufficiently for the claim?

Construct and transfer validity. Does the metric measure the stated construct, and what changes between simulation, hardware and human evaluation?

"Not established by inspected material" is not equivalent to "absent from the entire paper or codebase". This wording is particularly important for supplements, private evaluation logs and newer revisions that were not inspected. Likewise, an author's empirical success is not discredited simply because a different application needs stronger evidence; the review identifies the transfer boundary.

2.8 Synthesis rules and strength of gap claims

No meta-analysis of success rates is attempted. The corpus mixes different robots, information sources, trial lengths, task definitions and risk metrics. Quantitative values are used to explain a method's own evidence and its limits, not to form a cross-paper leaderboard. Thematic conclusions are instead built from converging mechanisms, explicit counterexamples and repeated methodological issues. For example, DWA and the biped MPC paper invalidate broad claims that dynamics or stopping were ignored; SCAND invalidates a claim that human judgments are absent from social-navigation evaluation [6, 16, 24].

A gap is described at one of three levels: a *demonstrated project deficit*, such as recorded contact failures; an *under-tested interface in this corpus*, such as calibrated observation-age/execution coupling; or a *candidate research contribution* requiring a more extensive nearest-neighbor literature search and a new mechanism. These levels must not be collapsed. The bounded corpus does not establish the absence of every delayed-CBF, robust/chance-constrained MPC, conformal-calibration or contact-aware manipulation method. Those literatures are explicit pre-submission search priorities.

The extraction and synthesis were prepared in a single AI-assisted review workflow, without independent second-person adjudication. The delivered structured records make such review possible but do not replace it. No literature-review protocol was preregistered, and no new physical trial was performed. These limitations define the strength of the conclusions while preserving a substantial increase in primary-material analysis over the previous abstract-led report.

3 The story begins: what makes navigation social?

3.1 From reaching a coordinate to sharing a space

The simplest navigation question is geometric: can a robot move from its present configuration to a goal without intersecting an obstacle? If the map and robot shape are fixed, a collision-free path is a sequence of configurations in free space. Even this question is not the whole control problem. A path does not specify how fast to traverse it, whether a motor can produce the requested acceleration, or whether the robot can stop before a previously unseen obstacle appears. DWA made this distinction explicit long before modern deep learning: searching reachable, stoppable velocities is different from selecting an appealing direction and converting it into motion afterward [16].

People introduce a second distinction. A person has an intention, a perception of the robot and the ability to react. A route that is physically possible may force someone to change direction, interrupt a group, block access or produce an uncomfortable encounter. Conversely, the most conservative geometric solution may stop indefinitely in the middle of the shared space. Thus neither "reaches the goal" nor "never moves into an occupied cell" is a complete definition of successful social navigation. The surveys organize this problem around human awareness, interaction context and the purpose of the task [25, 42].

The distinction is especially important for a delivery humanoid. Independent passage through a corridor, approaching a recipient and carrying an object past a seated person are different social situations. A distance acceptable during an invited handover may be inappropriate when overtaking an unaware pedestrian. The desired behavior therefore depends on the robot's task and the human's role, not only on relative coordinates. This is why the first research decision must be a task definition, rather than a choice between an optimization-based and a learning-based planner.

3.2 Five objectives that should not be collapsed

For the present review, the navigation problem has five distinct outcome families.

1. **Physical integrity:** avoid prohibited contacts, loss of balance, falls and dropped payloads. The relevant geometry includes whatever can hit a person, not necessarily just the base footprint.
2. **Task completion:** reach the intended goal and satisfy its stopping or delivery condition. Briefly entering a goal region is not necessarily a completed docking or placement action.
3. **Liveness and efficiency:** continue making useful progress, resolve temporary blockage and complete the task without excessive delay or intervention.
4. **Social appropriateness:** produce behavior that is understandable, contextually suitable and acceptable to affected people.
5. **Evidence validity:** establish that the sensors, scorer, timestamps and event definitions support the claimed outcome.

These outcomes can conflict. Increased clearance may lengthen a route; abrupt avoidance may protect base clearance while destabilizing a biped; early stopping may reduce the robot's motion into a person but increase the time it obstructs a passage. The evaluation guidelines consequently advocate a multidimensional account rather than one unqualified social score [17].

It is useful to distinguish a *proxy* from its target construct. Time within a specified personal-space radius is a geometric statistic. Perceived discomfort is a human response. The statistic may correlate with discomfort under a declared condition, but the relationship has to be tested. The proxemics literature makes the dependence on human, robot, environment and context particularly explicit [34]. The same caution applies to smoothness: low command jerk does not prove that the executed motion is smooth, and smooth motion does not prove that its social intent is legible.

3.3 A decision model that exposes the hidden assumptions

Let x_t denote the robot's physical state, z_t the surrounding agents and environment, and $o_{0:t}$ the available observation history. A planner chooses an action using a belief or estimate rather than complete knowledge:

$$u_t = \pi(b_t, g), \qquad b_t = \mathcal{E}(o_{0:t}), \qquad x_{t+1} = f(x_t, u_t, z_t, w_t), \tag{1}$$

where g is the task goal and w_t represents disturbances and modeling error. This notation is explanatory, not a new algorithm. It identifies four questions that recur throughout the

literature: what is observed, how uncertainty is represented, what action the policy actually commands, and how that action changes the physical state.

A planar simulator may make u_t the next velocity and realize it immediately. A humanoid navigation stack may instead make u_t a desired body velocity, then pass it through normalization, acceleration limits, a locomotion network, joint targets, PD actuation and contact dynamics. The physical transition is therefore mediated by a second policy and by execution state. Comparing these systems only on the label "velocity action" hides a major difference. SARL explicitly uses instantaneous velocity achievement; biped MPC explicitly models balance and terminal stopping; a LiDAR-to-motor policy integrates another part of the chain [6, 10, 48].

People's behavior introduces an additional conditional dependence:

$$p(z_{t+1:t+H} \mid o_{0:t}, u_{t:t+H-1}). \tag{2}$$

An independent constant-velocity forecast omits the dependence on robot action. Reciprocal avoidance imposes a particular response rule. An interaction-conditioned predictor tries to learn that dependence. None is appropriate merely because it is more complicated; the correct choice depends on whether the surrounding people actually observe and adapt to the robot and on the information available to infer their response [39, 41, 46].

3.4 What geometric methods solved, and what they left open

Velocity-space methods turned dynamic avoidance into a tractable geometric problem. DWA restricts the feasible command set using acceleration and stopping; velocity obstacles reason about relative velocities leading to future collision; ORCA distributes corrective responsibility among cooperating agents [15, 16, 46]. Timed trajectory optimization adds an explicit temporal structure to the local path [38]. These are not obsolete baselines simply because later methods learn their representations. They expose assumptions and failure reasons that are often harder to inspect in a neural policy.

Their limitations are specific rather than universal. A reciprocal guarantee does not apply unchanged to a person who never reacts. A safe velocity can become unsafe after post-hoc clamping. A finite-horizon collision-free trajectory may have no safe continuation. A local cost optimum may not escape a blocked route. The ORCA chapter itself recognizes the loss of its strict guarantee after kinematic/dynamic post-processing [46]. A serious comparison should use these limitations as experimental variables, not as a rhetorical reason to omit classical methods.

3.5 Why cooperation and freezing are two sides of the same problem

Suppose a robot predicts several plausible trajectories for every nearby person and refuses to enter any of them. As crowd density and prediction horizon grow, the union of those regions can cover every useful route. The robot freezes, even though people could pass each other by making small coordinated adaptations. Models of cooperation address this by reasoning about mutually compatible behavior [45]. Social-force and reciprocal models likewise generate collective organization from local interactions, although they encode different behavioral assumptions [21, 46].

The opposite failure is overconfidence in cooperation. A controller may select a path that is safe only if a pedestrian slows or steps aside. An inattentive person, mobility-aid user or person approaching from behind may not do so. The interesting research problem is therefore not to choose universally between cooperation and conservatism. It is to define which assumptions justify a particular action and what the robot does when the assumptions are uncertain or contradicted by observations. Nonreactive actors are useful for testing unilateral avoidance, but they cannot evaluate the quality of negotiated interaction.

3.6 Why learning changed the representation of the crowd

Learned navigation initially made it possible to amortize expensive interaction reasoning into a value function. CADRL represents local interactions in a robot-centric state; SARL learns how to aggregate several people; RGL learns relations and future state evolution; DS-RNN adds structured temporal reasoning [10–12, 28]. This progression is more informative than a simple "classical versus AI" division. The models move different parts of inference and planning into learned components and make different assumptions about the state available at deployment.

Learning also changes the experimental burden. A method's behavior depends on the training distribution, crowd policy, reward, privileged inputs and model selection procedure. A planner trained with circle-crossing ORCA agents can learn useful anticipation while remaining poorly characterized under detector dropout or a different walking controller. Conversely, a failure after naive transfer may measure an interface mismatch rather than an inherent weakness of the learning architecture. Equal input information and comparable tuning budgets are prerequisites for an interpretable comparison.

3.7 Prediction is an input to a decision, not the final objective

Social LSTM, Social GAN and Trajectron++ demonstrate increasingly expressive representations of pedestrian futures: interaction pooling, multimodal sampling, heterogeneous data and dynamics/ego-plan conditioning [1, 19, 41]. Yet a robot must decide before knowing which future will occur. A predictor's best sample can be close to the recorded trajectory while the distribution assigns too little probability to a dangerous alternative. Conversely, a simple short-horizon estimate can support an effective decision when observations are frequent and the robot has time to brake.

This motivates a decision-centered evaluation: which errors change the chosen action, whether the predicted occupied region covers the actual person at the encounter, and whether the resulting robot trajectory is executable. Lower average displacement error is useful evidence about a forecasting task, but is not by itself evidence of fewer contacts or better social behavior. The later sections use this distinction to connect perception, timing and control.

3.8 Embodiment closes the loop

A biped must remain balanced while it avoids people. The 2016 biped crowd paper already unifies balance, stopping and perception uncertainty through viability and lexicographic MPC [6]. Learned locomotion later broadened the achievable repertoire through adaptation, perceptive control and capability-aware navigation [22, 26, 27, 33]. Humanoid imitation methods show how arm objectives, posture and dynamics mismatch alter the control problem [13, 20]. These developments explain why a planar command model cannot simply be assumed correct for a humanoid.

They also constrain novelty. "Account for dynamics" and "combine navigation with locomotion" are already established research directions. What can remain open is a particular, validated interface: for example, how a measured response distribution for a deployed policy should combine with observation age and person uncertainty when selecting an avoidance action. The contribution must be the specified model, decision rule or causal result, rather than the fact that the robot has legs.

3.9 From a successful maneuver to dependable social autonomy

Recent LiDAR-conditioned policies and visual generative navigation systems already demonstrate sensor-driven avoidance and transfer to legged or humanoid platforms [18, 47, 48]. Benchmarks now include grounded human trajectories, interactive avatars, diverse urban scenes and composed tasks [5, 14, 37, 43, 49]. A research programme must therefore ask why successful components still fail when composed, rather than assuming that no one has integrated them before.

The central story of this review is that **the field progressively exposes more of the closed loop**. Geometry makes collision conditions explicit; interaction models make human response explicit; learning makes richer behavior representations practical; perception makes partial observability unavoidable; embodied control makes command execution conditional; and complete-mission evaluation exposes failure propagation across all of them. The proposed novelty roadmap follows that story: establish a measured baseline, define a testable new interface or mechanism, isolate its effect and demonstrate that it matters under conditions the nearest prior methods do not already resolve.

4 The current work and its evidence boundary

This section retains the original repository-inspection context from 22 September 2026 so that its numerical observations remain attributable. The later published development baseline and current priorities are linked from the project status page. The retained v84 full matrix reports 10/18 strict passes, 46/54 goals and six contacts; the payload report documents two consecutive isolated zero-person fixture completions. These later records do not establish navigation or crowd transport acceptance, and no new robot trial was run for this review.

4.1 Architecture relevant to the review

Geoff-Sim is a simulation-first Unitree G1 research platform. Its documented pipeline connects MuJoCo/DDS robot state and simulated sensors to ROS 2, SLAM Toolbox, Nav2, dynamic-person tracking, short-horizon prediction, human-aware command selection, a collision supervisor, and a learned locomotion controller. This is an appropriate architecture for investigating the interfaces between autonomy layers, rather than only isolated planner performance.

The repository inspection covered `README.md`, `docs/ONBOARD_SOCIAL_NAVIGATION.md`, `docs/DYNAMIC_ROUTE_PLANNING.md`, `docs/BENCHMARK_EXPERIMENTS.md`, and the implementation `motion_envelope.py` in the bridge package. File hashes and the repository revision/dirty-state context are retained in `project_evidence.json`. The working tree contained ongoing development; the reviewed files, not the base commit alone, define this assessment's project snapshot.

4.2 Oracle and sensor-only modes are different experiments

The oracle social-waypoint mode uses simulator-derived person tracks and a known static map. The onboard mode uses local-frame raycast LiDAR, IMU and joint information, scan-registration odometry, online SLAM, observation-based tracking, and constant-velocity pedestrian forecasts. Evaluation truth remains available for scoring but is excluded from the documented navigation path. The source audit is a graph/configuration check rather than a security sandbox. This distinction is scientifically important: privilege can enter through localization, goal selection, maps, or tracks, not only through a planner's direct subscriptions.

The onboard detector is geometric, not a semantic camera or learned person classifier. The odometry front end assumes approximately planar walking, and its covariance is not a calibrated localization uncertainty model. Consequently, the present system provides an experimental platform for perception-aware navigation, rather than a completed solution to open-world person recognition or general uneven-terrain localization.

4.3 What the recorded trials establish

The onboard status document explicitly states that an earlier oracle 24/24 result has not been reliably reproduced with onboard sensing. Development trials include an eight-goal completion and several collisions and timeouts. These runs span implementation changes and must not be pooled into a single algorithm success rate. The dynamic-route validation records 136 route updates but only three of four goals and one collision. This establishes active integration, not reliable avoidance.

Timing records are similarly informative but conditional. One onboard trial reports planner p95 47.115 ms, tracker p95 53.501 ms, and track age at output p95 0.305 s. A later route-validation trial reports request-to-matching-route p95 2.649 s, including retries and blocked-route periods. These quantities measure different things: node computation time, observation age, and delivered route latency cannot be substituted for one another. None is a hard real-time guarantee. The reported 45 ms candidate-search budget can overrun.

4.4 The key technical opening

The shared motion envelope already normalizes command bounds and models acceleration-limited, delayed planar motion. Its source explicitly states that the defaults are *conservative engineering assumptions, not a fitted G1 response certificate*. That admission identifies a tractable research step: measure the command-to-motion relationship and test whether a calibrated model changes safety decisions. Current velocity caps vary by profile, so experiments must freeze the policy, gains, normalization, delay model, and active command limits instead of quoting one cap as universal.

The batch tools already preserve failed/invalid/missing trials and distinguish control-source evidence from scoring truth. These are useful foundations, but their presence does not by itself validate the response model or prove a novel navigation method.

5 Classical and learning-based social navigation

5.1 Geometric and dynamically constrained avoidance

The Dynamic Window Approach (DWA) searches feasible velocities subject to the robot's dynamics and stopping considerations [16]. Velocity obstacles (VO) reason about relative motion and select velocities outside collision-producing regions; the original work also intersects these with acceleration-admissible velocities [15]. Therefore, merely adding an acceleration limit to a geometric planner is not a sufficient novelty claim. Their continuing value is as interpretable baselines: failures can be traced to motion estimates, admissible sets, or objective functions.

Timed trajectory optimization offers another classical comparator. Sparse trajectory optimization underlying the timed-elastic-band family jointly adjusts a path and its temporal structure [38]. It makes time and kinematic constraints explicit, but practical behavior still depends on initialization, obstacle representations, and the fidelity of its executable motion model. A G1 can have command deadbands, axis coupling, turning-arc restrictions, and balance-related transients that are not captured by a generic planar robot model.

ORCA distributes collision-avoidance responsibility between interacting agents [46]. This is powerful when agents conform sufficiently to the reciprocal assumptions. The repository's independently walking actors do not react to the robot, so an unmodified reciprocal assumption can be misleading. An ORCA comparison should declare whether people participate in the same avoidance rule, remain nonreactive, or follow recorded trajectories. A planner baseline and a pedestrian simulator are different uses of ORCA and should not be conflated.

Research implication. The useful distinction is not "dynamics-aware versus all previous kinematic work". It is whether a particular model of the deployed locomotion policy, command filtering, and observation delay improves decisions beyond established feasible-velocity and timed-trajectory methods.

5.2 Interaction, reciprocity, and freezing

Social-force modeling provides a compact account of interactions among pedestrians [21]. Its interpretability makes it useful for simulation and baseline reasoning. It does not make a simulated crowd a validated model of all human behavior, particularly for groups, distraction, or unexpected encounters with a humanoid. Crowd generation should therefore be an experimental factor rather than an unquestioned ground truth.

Trautman and colleagues explicitly study cooperation in dense crowds [45]. The central conceptual challenge is that treating every possible human motion as an obstacle can prevent progress, while assuming cooperation can be unsafe when the assumption is wrong. The same tension appears in Geoff-Sim: stronger margins may prevent a local encounter but produce waypoint timeouts; a rapid command can improve progress while increasing collision or fall risk. Performance should be evaluated on a collision–completion frontier rather than a single "safety score".

The early human-aware-navigation survey and subsequent taxonomies emphasize that obstacle avoidance is only one part of navigation among people [25, 42]. A robot must also make its behavior legible, respect interaction context, and avoid disrupting people. A successful nonreactive-actor benchmark validates unilateral avoidance under a defined motion model. It does not measure cooperative negotiation or human comfort.

5.3 Learned interaction models

CADRL learns a value function for decentralized collision avoidance, moving much of the interaction computation into offline training [12]. SARL uses attention to reason about collective human interactions rather than only independent robot–human pairs [10]. Relational Graph Learning adds graph-structured representations and predictive lookahead [11]. DS-RNN incorporates spatial and temporal relationships and addresses partial observability; its abstract also reports transfer to a TurtleBot 2i [28]. These methods provide a useful progression from pairwise local reasoning to temporal crowd representations.

The critical comparison is the information and dynamics contract. A policy consuming tracked positions and velocities does not perform the detector's task. A benchmark using a circular planar robot does not necessarily test balance, upper-body swing, or actual stopping transients. Conversely, it would be inaccurate to label all learning-based methods "oracle-only" or "simulation-only", particularly given DS-RNN and newer sensor-level systems. Each comparator needs its own documented observation interface.

The 2025 review and benchmarking paper identifies strengths of learning-based and end-to-end approaches in coordination scenarios [2]. This supports including a learned navigation

comparator if resources permit. However, importing a policy trained with different pedestrian reactions, speed ranges, sensors, or reward conventions can create a distribution-shift test rather than a fair algorithm comparison. Such transfer is valuable, but should be labeled and analyzed separately from matched training.

Research implication. A modular response-aware approach is attractive for interpretability and data efficiency, not because learned policies cannot avoid people. Its case depends on comparable input information and a demonstrated advantage under delay, missed detections, or changed locomotion response.

6 Prediction and onboard perception

6.1 From point forecasts to interaction-conditioned distributions

Social LSTM introduced learned interaction-aware trajectory prediction; Social GAN explicitly represents diverse socially plausible futures; Trajectron++ combines graph structure, heterogeneous agent data, and dynamical feasibility [1, 19, 41]. Trajectron++ can condition predictions on an ego-agent plan, which is especially relevant when robot action influences pedestrian behavior. Rudenko et al.'s survey provides a broad taxonomy of trajectory modeling and contextual information [39].

These developments challenge a single constant-velocity forecast over long horizons, but do not imply that a larger network will improve this project's closed-loop behavior. Short-horizon constant velocity is a valuable baseline: it is inexpensive, auditable, and can be more stable than acceleration extrapolation from noisy surface centroids. The onboard profile's use of constant velocity is therefore defensible as a starting point. Its limits must be measured at crossings, turns, stops, and occlusion exits.

Prediction error averaged over entire trajectories can conceal the errors that cause a collision. Multi-sample best-of-K displacement error can also reward producing one good sample without providing calibrated probabilities. For planning, the relevant questions are whether the occupied region covers the person at encounter time, whether probability estimates are calibrated, and whether the chosen robot motion remains feasible. This is a synthesis of the planning problem, not a claim that all reviewed predictors omit these issues.

For a track observed at t_o and used at t_c, define its age $a = t_c - t_o$. A minimal time-aligned mean prediction is

$$\hat{p}_h(t_c + \tau) = \hat{p}_h(t_o) + \hat{v}_h(t_o)(a + \tau). \tag{3}$$

The corresponding occupied region must also widen with measurement, velocity, and unmodeled-motion uncertainty. Equation 3 is only a proposed baseline formulation; it is not evidence that the current implementation has calibrated coverage. Repeated publication of an old track must not reset t_o.

6.2 Detection and tracking are not solved by geometry alone

DROW establishes learned detection from sparse two-dimensional range data, initially emphasizing wheelchairs [4]. DR-SPAAM uses spatial attention and recurrent feature updates for person detection in range scans [23]. These are pertinent perception baselines, but transferring their assumptions to a moving torso-mounted 3D LiDAR requires an explicit projection and domain-adaptation study. A flat scan through legs is not equivalent to clipped, oscillating torso returns.

JRDB provides egocentric human detection/tracking data in built environments [32]. SCAND provides human-demonstrated socially compliant navigation on morphologically different platforms, including a quadruped and a wheeled robot [24]. The former is useful for characterizing missed detections and association; the latter is useful for behavioral imitation and judging which maneuvers humans demonstrate. Neither automatically supplies the G1-specific LiDAR mounting, sway pattern, or command response of this project.

The repository already attempts partial-body track retention and measured fragment recovery. These are practical responses to a genuine problem: an observed surface centroid is not a stable estimate of a person's body center. Differentiating that centroid may produce an apparent acceleration that belongs to visibility geometry rather than human motion. Ground-truth evaluation should separate center bias, velocity error, identity continuity, track age, and complete misses. The current nearest-track diagnostic is explicitly not identity-matched detection recall; it should not be reported as such.

6.3 Localization, mapping, and the moving sensor

FAST-LIO2 demonstrates direct LiDAR–inertial estimation, while ORB-SLAM3 provides a visual/visual–inertial alternative [8, 50]. They identify strong adjacent baselines and underline that state estimation is a developed field. The project's planar scan-registration front end should be positioned as an engineering component unless a new estimation method is separately evaluated.

Walking creates a feedback loop between body motion, sensing, and planning. Attitude errors can affect ground removal and scan projection; localization drift can shift both maps and tracks; dynamic-person masking can remove useful geometry. Correctly handling unknown map cells does not by itself resolve these effects. Conversely, adding a more capable estimator may not fix collision failures caused by stale tracks or unmodeled braking.

Research implication. A controlled oracle-to-onboard decomposition is stronger than a single end-to-end success number. Independently substitute estimated pose, observed tracks, and online maps to locate which uncertainty source changes decisions. Then test interactions between those sources rather than assuming their effects add linearly.

7 Legged control and the closest humanoid competitors

7.1 Locomotion competence versus navigation competence

Massively parallel training reduces the time needed to learn locomotion policies [40]. RMA learns rapid adaptation to changing locomotion conditions [26]. Robust perceptive locomotion combines exteroception and proprioception to handle difficult terrain [33]. These advances make a learned low-level policy an increasingly practical component, but their task objectives differ from mission-level navigation through a crowd. Good velocity tracking in a locomotion benchmark does not specify a safe pedestrian-avoidance trajectory.

DeepMimic establishes example-guided physics-based motion imitation [35]; ExBody transfers expressive whole-body behavior to humanoids while relaxing lower-body imitation to support velocity tracking [13]. These are relevant to natural motion and potentially legibility, but motion resemblance is not evidence of perceived social comfort. Human evaluation would be needed to establish that relationship.

The boundary between locomotion and navigation is already an active research area. ANYmal Parkour uses a hierarchy that accounts for locomotion skills when choosing how to traverse

obstacles [22]. The wheeled-legged navigation work integrates adaptive locomotion, mobility-aware local navigation, and large-scale planning, reporting kilometer-scale missions [27]. Earlier work explicitly addresses safe biped navigation in crowds [6]. These are direct counterexamples to an overly broad claim that navigation has always ignored legged execution.

7.2 ASAP: why nominal simulator dynamics are not enough

ASAP pretrains motion-tracking policies, collects real-world trajectories, learns a residual action model to align simulated response, and fine-tunes the policy in the corrected simulator [20]. The inspected version compares transfer between simulators and to a real G1. Its tables examine both open-loop dynamics mismatch and closed-loop motion imitation. This separation is important: fitting transitions and improving a deployed policy are related but distinct achievements.

For Geoff-Sim, ASAP strengthens the motivation to calibrate response rather than trust nominal gains. It also limits the novelty of "learning dynamics corrections". The proposed contribution here would operate at a different interface: a short-horizon reachable motion envelope for a frozen locomotion policy, used to select or veto navigation commands. It would not duplicate ASAP's whole-body policy fine-tuning objective. Dynamics randomization is another established transfer tool [36], but randomized training does not supply a calibrated deployment-time stopping bound by itself.

ASAP is not a crowd-navigation evaluation. Its imitation errors and skill success rates cannot be numerically compared with pedestrian collision or waypoint completion rates. The useful transfer is methodological: evaluate model fidelity and downstream behavior separately, on held-out conditions.

7.3 LiDAR-conditioned safe and comfortable humanoid locomotion

The most directly overlapping 2025 work is *End-to-End Humanoid Robot Safe and Comfortable Locomotion Policy* [48]. It combines LiDAR features and proprioceptive history, uses a constrained Markov decision process, translates discrete barrier-function ideas into training costs, and incorporates comfort-oriented rewards. The inspected paper reports Unitree G1/Livox Mid-360 deployment and onboard execution. Its actor/critic description explicitly distinguishes privileged training information from deployment inputs.

This prior work makes "first sensor-driven safe humanoid avoidance" an indefensible positioning. It also offers a useful contrast between an end-to-end learned controller and a modular planner plus supervisor. However, several details constrain how strongly its results can be used:

- Table III reports 30 trials per scenario and rounded success values of 86% for the proposed method versus 70% for P3O in the dynamic-agent case. The same proposed method reports 60% in a narrow passage. These are scenario-dependent outcomes, not universal avoidance guarantees.
- Table II reports nonzero unsafe-region time even for the proposed method. The training discussion uses a 0.8 m safety distance, while that table defines unsafe space below 0.6 m. The evaluation thresholds must be kept separate rather than merged into a single certified clearance claim.
- The described barrier condition is converted into an optimization cost during learning, rather than enforced as an online action constraint. A theoretical invariant-set condition is not automatically inherited by an approximately trained policy under sensing and dynamics errors.
- Comfort is operationalized through reward terms and geometric/dynamic proxies. The inspected experiments do not establish perceived comfort through a reported participant

study. This leaves room for later HRI validation, but not a claim that comfort was wholly absent from prior work.

These observations are not a re-evaluation of the paper's code or raw data. They identify which claims its printed evidence can support and which quantities would need harmonization in a comparative experiment.

Omni-Perception is another close 2025 comparator. The expanded reading of its v2 manuscript examines proximal/distal point-cloud processing, recurrent features, sensor rendering, ablations and physical scenarios [47]. Its architecture prevents novelty from being assigned merely to direct point-cloud perception or multi-simulator raycasting. Its reported success and collision percentages require careful event interpretation; Section 10 and its detailed appraisal explain why they cannot be read as complementary outcomes.

7.4 FLUX: a 2026 cross-embodiment navigation competitor

FLUX uses rectified-flow trajectory generation followed by dynamic-scene reinforcement-learning refinement, and introduces DynBench [18]. The inspected version includes static and dynamic navigation tasks and reports transfer to wheeled, quadrupedal, and G1 humanoid platforms. Its deployment uses an RGB-D camera and inference on a remote RTX 4060 machine. This matters when comparing onboard compute constraints and communication latency with Geoff-Sim.

Three details are especially relevant to critical positioning:

1. Its Table II reports SocialNav success of 64.0 versus 59.3 for NavDP. This is meaningful within that protocol, but goals, episode termination, pedestrian speeds, and social-distance definitions differ from Geoff-Sim. It is not a directly comparable headline improvement.
2. Section V defines "Collision Count" as intimate-space intrusions below 0.45 m, a proxy for collision risk. Later prose describes values with percent signs while the metric definition describes counts. Accordingly this review does not reinterpret those values as physical contact probabilities. Physical contacts and proxemic intrusions should be distinct outputs in the proposed study.
3. Table IV reports 216.5 ms inference for the selected six-step configuration. This is a different timing boundary and compute setup from the project's local-planner p95 measurements. It motivates measuring end-to-end observation-to-actuation age; it does not prove one system is faster under matched conditions.

FLUX's limitation discussion acknowledges missing explicit social reasoning and predominantly simulated indoor pedestrian evaluations. Its existence still decisively shifts the research gap: cross-embodiment dynamic navigation is no longer an untouched area. A narrowly scoped G1 response and uncertainty study can be complementary to such general policies.

8 Safety, middleware, and whole-system timing

8.1 Filtering commands versus guaranteeing safety

Control barrier functions provide a formal route to invariant-set conditions and quadratic-program safety filtering under stated model assumptions [3]. The safe-learning survey distinguishes empirical robustness, learning-based improvement, and formal certification [7]. These distinctions should govern terminology in this project. A collision supervisor that checks predicted rollouts is a safety-oriented mechanism; without validated disturbance bounds and recursive feasibility, it is not automatically a safety certificate.

A navigation command u is transformed before its physical effect appears. Let $\mathcal{N}$ denote

normalization, deadbands, and command shaping, and let $\hat{f}_\theta$ approximate policy-level motion response. A candidate rollout can be expressed as

$$\hat{x}_{k+1} = \hat{f}_\theta(\hat{x}_k, \mathcal{N}(u_k), d_k), \tag{4}$$

where d_k includes the measured or modeled delay. This notation summarizes a proposed experimental model, not a claimed full-order description of G1. Its accuracy must be measured during braking, turning, and combined-axis motion, not only steady straight walking.

For a one-dimensional stop with constant speed during delay and constant deceleration afterward, an illustrative distance is

$$d_{\text{stop}} = vd + \frac{v^2}{2a_{\text{brake}}}, \qquad a_{\text{brake}} > 0. \tag{5}$$

The equation explains why zero commanded velocity is not zero stopping distance. It is not a bound until delay and achievable braking have been validated, and it omits a moving pedestrian's continuing approach. For example, if a person approaches at 1 m/s, an observation age of 0.305 s corresponds to 0.305 m of unobserved motion before future-horizon prediction even begins. This is an illustrative calculation using a chosen human speed, not a reconstructed collision from the repository.

8.2 Planner–supervisor interaction

A planner may choose a trajectory that the supervisor repeatedly vetoes. Even when every individual veto seems prudent, repeated disagreement can cause oscillation or starvation. A suitable study measures override rate, rejected-command reasons, clearance under the actual applied command, yield duration, and progress after resumption. It should also distinguish stopping in front of an approaching person from moving to a genuinely lower-risk configuration: stationary is not synonymous with safe in a dynamic environment.

For a humanoid, emergency velocity changes can create a second hazard: instability or contact by a swinging body segment. A planar footprint can be useful if conservatively validated, but it does not directly measure full-body swept volume, fall reach, or contact force. Such distinctions are particularly relevant to the project's use of independent pedestrian actors.

8.3 ROS 2, Nav2, and SLAM are enabling infrastructure

ROS 2 provides the communication and lifecycle architecture [31]. Navigation2's behavior-tree orchestration is described in The Marathon 2 [30], and SLAM Toolbox supports the mapping component [29]. MuJoCo provides the physics foundation [44]. Combining these tools is valuable research engineering, but their presence is not a new navigation algorithm. Nor does middleware design establish hard real-time performance for an entire application.

The current event-driven route monitor, forecast costmap layer, and retained path execution offer a specific timing experiment: when does event-triggered replanning improve collision-free progress relative to fixed-rate replanning under the same compute budget? A 5 Hz request limit is not a 200 ms route delivery guarantee. Queueing, costmap availability, retries, and action transport belong in the latency measurement. Global predictive costs and local collision checking should also be ablated independently to avoid crediting the same forecast twice.

9 Benchmarking and social validity

9.1 Existing benchmarks already cover important parts of the problem

SocNavBench provides grounded scenarios, consistent metrics, and a common testing framework [5]. SocialGym 2.0 emphasizes configurable multi-robot navigation in shared human spaces [9]. Habitat 3.0 adds human avatars, human-in-the-loop interaction, and collaborative tasks [37]. MetaUrban extends compositional urban environments and heterogeneous mobility evaluation [49]. These address realism, comparability, interaction, or scale that a small custom plaza cannot assume.

OmniNavBench is particularly important to a 2026 gap analysis [43]. Its inspected version includes 170 environments and 1,779 human-teleoperated trajectories, multiple embodiments including Unitree H1, continuous physical stepping, and composed navigation tasks. The low complete-task success of several evaluated models highlights a distinction between individual skills and reliable task sequences. Its social metric uses a personal-space threshold, which should not be equated with a full-body contact sensor. Geoff-Sim's proposed advantage should therefore be fine-grained control/measurement provenance and calibrated execution analysis, rather than the first physically embodied multi-task benchmark.

The very recent DPed-VLN manuscript describes language navigation with dynamic pedestrians, paired instructions, and social-safety metrics [14]. It is adjacent rather than a direct baseline for a LiDAR/Nav2 waypoint stack. Its inclusion demonstrates why "navigation in dynamic pedestrian scenes" is too broad a novelty claim. The expanded reading includes its protocol and tables, which expose different success–risk tradeoffs across imitation and reinforcement learning. Humanoid pedestrian avatars are not evidence of a low-level humanoid robot controller; that distinction remains explicit.

9.2 Safety is necessary but not sufficient for social navigation

The evaluation guidelines articulate safety, comfort, legibility, politeness, social competency, agent understanding, proactivity, and context responsiveness [17]. The 2026 proxemics taxonomy further emphasizes human, robot, environment, and contextual influences [34]. A fixed circular personal-space penalty is consequently a controllable proxy, not a universal measure of comfort.

A simulator can validly measure minimum body clearance, contact events, time in specified proxemic regions, approach speed, jerk, and route progress. To infer perceived safety or comfort, the study needs appropriately designed human judgments, such as counterbalanced video ratings or later controlled in-person comparisons. These should separate morphology, speed, yielding style, and context. A humanoid's natural-looking gait may improve legibility or may increase expectations that the controller does not meet; the direction of the effect is an empirical question.

9.3 A useful validity hierarchy

1. **Mock regression:** verifies scenario, scoring, aggregation, and failure-handling mechanics; it provides no physical avoidance evidence.
2. **Oracle physics:** tests planning and execution with privileged perception; it cannot establish sensor-only performance.
3. **Onboard-sensor simulation:** tests the integrated observation and execution pipeline under the simulator's sensing/physics assumptions.
4. **Cross-simulator or perturbed-model evaluation:** probes sensitivity to selected modeling assumptions; it is not itself hardware transfer.
5. **Hardware and human evaluation:** addresses physical transfer and human-centered outcomes under an explicit deployment protocol.

The current project spans the first three levels with incomplete reliability at level three. Reporting the hierarchy avoids overstating either the mock tests or the passing oracle demonstrations.

10 Deeper thematic synthesis: where the methods agree and disagree

10.1 The action interface is the hidden organizing variable

The most consequential difference between several reviewed methods is not whether they use a neural network. It is the meaning of the action they evaluate. DWA evaluates planar velocities constrained by reachable and stoppable motion; ORCA selects a collision-avoiding velocity under a simplified reciprocal model; SARL's formulation makes the selected velocity instantaneous. Biped MPC instead includes a balance-compatible stopping condition. At another level, a LiDAR-driven locomotion policy emits joint commands, while a visual navigation model may emit waypoints or trajectory-like targets for a separate controller [6, 10, 16, 18, 46, 48].

This difference changes the problem being solved. Let $\mathcal{A}_{\text{plan}}$ be the planner's nominal action set and $\mathcal{T}_{\text{exec}}(u, x)$ the distribution of physical trajectories produced by executing u from state x. A geometric planner effectively substitutes a deterministic nominal trajectory for this distribution. A useful response-aware method must improve that substitution in a way that affects decisions, not merely produce a better-looking predicted path. In particular, the map from command to motion must include whatever normalization, saturation, deadband, slew and policy history is active in the deployed interface.

This framing also explains why post-hoc filtering is not innocuous. If a planner checks a curved forward command and a supervisor replaces it with a stop, the executed stopping trajectory may have different clearance from a moving person. If a policy then produces lateral drift, neither previously checked path is the real trajectory. A common command vocabulary is insufficient; the layers need a common interpretation of the command's short-horizon physical effect. The ORCA limitations and wheeled-legged failure analysis already motivate this concern, so the novel object must be a specific shared model or coordination rule [27, 46].

10.2 Reciprocity is an intervention, not a harmless simulator setting

SARL's visible and invisible settings provide a particularly clear contrast. The same general navigation problem produces very different ORCA performance depending on whether the surrounding agents react to the robot. Its Table I reports success 0.43 for ORCA in the invisible setting, compared with 0.99 in the visible setting in Table II; these are distinct protocols rather than a single algorithm failure rate [10]. SocNavBench explicitly changes collaboration coefficients for nonreactive pedestrians and describes the kinematic choices used in baseline adaptation [5].

The experimental implication is stronger than "use more realistic crowds". Reciprocal, independent and replayed actors answer different counterfactual questions. Replay holds human trajectories fixed and can isolate robot-side changes, but it removes the human response that an interactive planner is meant to exploit. Reciprocal simulation permits that response but may make it too predictable or overly accommodating. A data-driven actor can improve behavioral diversity while introducing another learned model whose errors need evaluation. One should therefore report results by crowd regime and avoid pooling them into a universal social-navigation success rate.

For Geoff-Sim, the nonreactive regime is scientifically valuable precisely because it tests unilateral responsibility. A contribution in that regime should be named accordingly. Claims

about cooperation, politeness or negotiation require additional conditions in which the human can respond and the effects on the human are actually measured [17, 25, 34].

10.3 A better representation can conceal a different experiment

The transition from CADRL to SARL, RGL and DS-RNN is often summarized as a sequence of more expressive networks. A closer reading shows simultaneous changes in aggregation, dynamics modeling, memory, training and observation conditions. RGL's comparison explicitly replaces an optimistic next-state assumption with a linear model before attributing gains to relational prediction. DS-RNN introduces a recurrent-attention ablation with matched PPO settings, as well as limited-view and group conditions [10–12, 28]. These decisions are part of the contribution's evidential strength, not implementation trivia.

A proposed response-aware system must meet the same standard. If it receives fresher tracks, a more capable policy, more CPU time and a different supervisor at once, a higher completion rate cannot be assigned uniquely to calibration. Conversely, running an externally trained baseline through an incompatible action adapter can manufacture a failure. The comparison contract should list inputs, predictor, action space, control rate, training data, runtime hardware, post-processing and evaluator for every arm. A controlled ablation changes one mechanism or uses a design that estimates interactions explicitly.

10.4 Prediction quality has at least three distinct meanings

First, a forecast can fit the recorded trajectory under an offline metric. Second, it can represent uncertainty with useful probability or coverage. Third, it can improve the decision made by the robot. Social LSTM, Social GAN and Trajectron++ address different parts of this progression [1, 19, 41]. They cannot be collapsed into a claim that any learned prediction is automatically safer than constant velocity.

Two details illustrate the danger. Social LSTM's Table 1 includes a privileged destination input for IGP, and its aggregate final-displacement entry is not better than every comparator. Social GAN evaluates the best of several samples and explicitly discusses differences from the original Social LSTM training setup. Trajectron++ adds distributional and dynamics-related metrics, including KDE NLL, but probability calibration under deployment shift remains a separate question. A method can generate a plausible future without assigning the right probability to a collision-critical alternative.

For a practical planner, the relevant quantity is not just $\|\hat{p}_h - p_h\|$ averaged over a test set. It is whether the predicted occupied region $\mathcal{O}_h(\tau)$ contains the person at the times when candidate robot trajectories approach them. Coverage should be examined by horizon, occlusion, maneuver, range and observation age. A broad region can increase coverage while making every route infeasible; the associated liveness cost must be measured. This leads to a joint occupancy–execution problem rather than a competition based on a single forecasting score.

10.5 The 2016 biped paper changes the novelty boundary substantially

The expanded primary reading of Bohorquez et al. [6] is especially important. Its formulation already includes center-of-mass/center-of-pressure dynamics, foot-placement feasibility, a terminal capture/stopping condition and uncertainty-expanded pedestrian occupancy. In its notation, a separation margin can grow with position error and velocity error over time. That is direct prior art against a claim that combining biped dynamics, uncertainty and stopping is new.

The paper also distinguishes passive safety from actually preventing every collision. Its numerical people do not avoid the robot, and the study varies field of view, horizon, crowd speed

and uncertainty. A longer preview is not uniformly helpful when the sensor cannot observe the relevant closing region. Most strikingly, one relaxation reduces collisions partly by allowing infeasible robot motion and falls; another preserves dynamic feasibility and obtains a better total-failure tradeoff. This is a concrete demonstration that optimizing one safety proxy can transfer risk to another outcome.

The defensible extension is therefore narrower: identify and validate the short-horizon response of a deployed learned locomotion policy, connect it to acquisition-time uncertainty under real onboard perception, and test its value relative to a generic uncertainty-aware/biped-feasibility model. The distinction between a reduced-order analytic walking model and an empirically calibrated policy interface could matter, but it is an experimental hypothesis, not a novelty result established by changing robot names.

10.6 Read recent end-to-end safety claims at the level of constraints

The safe-humanoid paper presents a formal definition of safe behavior and a barrier-inspired training mechanism. These are not the same object. Its text states that the barrier violation is used as a cost rather than as an online action constraint [48]. Approximate optimization of an expected discounted cost does not directly provide the pointwise invariance property of an online feasible CBF controller [3, 7]. This distinction is compatible with useful empirical performance; it simply prevents an unjustified guarantee from being carried across interfaces.

The tables likewise support a qualified interpretation. The proposed method has better dynamic-agent and narrow-passage results in the reported comparison, but not the best suspended-obstacle result. Nonzero unsafe-region duration remains, and the evaluation and training distance thresholds differ. The ablation prose also makes it difficult to isolate every contribution of the barrier-based cost from the comfort-related rewards without a fully factorial comparison. These are specific reasons to request matched variants and definitions, rather than to dismiss constrained RL as a family.

Omni-Perception adds another lesson. Its PD-RiskNet ablation reports both 76.7% success and 56.7% collision for one method; those are evidently not complementary exclusive outcome categories in the printed table [47]. Without the event semantics, success cannot be reinterpreted as collision-free completion. The paper's qualitative sensor comparison, rendering throughput and real-world scenario table are also separate evidence products. They should remain separate in an analysis of sensing fidelity, computational efficiency and locomotion reliability.

10.7 General-purpose navigation broadens the task but not automatically the guarantee

FLUX, OmniNavBench and DPed-VLN extend the field toward dynamic scenes, cross-embodiment deployment and composed or language-conditioned tasks [14, 18, 43]. They demonstrate that the ambition of the project cannot be positioned as the first movement beyond isolated point avoidance. Their evaluation contracts, however, differ substantially.

FLUX defines a collision-related statistic as intimate-space intrusion and uses task termination based on progress, timeout and goal proximity. Its inference timing and remote deployment setup are not comparable with a local-planner callback p95. OmniNavBench distinguishes subtask predicates, complete-episode success and personal-space intrusion over time; strong partial progress can coexist with poor whole-mission success. DPed-VLN shows that a method can improve goal-reaching while worsening a human-collision metric, and that adding moving pedestrians can change risk much more than it changes success.

These comparisons motivate an explicit metric ontology: geometric overlap, native contact, intimate-space intrusion, personal-space duration, failure to stop, fall and incomplete mission are

separate events. A common name such as "collision rate" does not make the events equivalent. A future benchmark contribution should publish the event predicates and time basis so that readers can reconstruct exactly what was measured.

10.8 Perception and execution uncertainty interact rather than simply add

DROW and DR-SPAAM show that surface measurements need normalization and learned aggregation to estimate an object's center; JRDB shows that misses, localization error and identity continuity can have different rankings [4, 23, 32]. FAST-LIO2 and ORB-SLAM3 show that robot-state estimation itself has a sensor, timing and calibration contract [8, 50]. A pedestrian position transformed through an erroneous robot pose is not an independent error source from the robot's estimated route.

An uncertainty-aware navigation study should therefore separate measurement noise, systematic center bias, unobserved motion, state-estimation error and command-response residuals. A single isotropic margin can conceal these mechanisms. Nonetheless, adding a separate learned distribution for every error source may make inference slow enough to worsen the total problem. The objective is not maximal model complexity, but a representation that changes the right decisions at the available observation age and compute budget.

This is why a source-audited factorial experiment is valuable. Replace one estimated component at a time with its evaluation counterpart, then test interactions between factors. If a better response model helps only when person observations are time-aligned, that interaction is a substantive result. If it does not help even with oracle perception, the main limitation may be the controller or decision rule rather than the detector.

10.9 Embodied adaptation makes the "same controller" a qualified statement

RMA, perceptive locomotion, ExBody and ASAP demonstrate different ways in which experience, sensor belief, posture and dynamics alignment affect physical behavior [13, 20, 26, 33]. A network file and a maximum speed do not fully identify the executed dynamics. Observation history, command normalization, PD gains, arm authority, payload and reset state matter as well. The wheeled-legged study's tracking and memory failures reinforce that this is already an active systems-level research problem [27].

For social locomomanipulation, a reaching arm changes both dynamics and collision geometry. A base-only envelope may be a useful approximation, but it requires validation across arm poses and load transitions. First measure where the approximation changes collision decisions. Only then is the extra cost of articulated swept-volume checking justified. A failure of the simpler model can be a meaningful finding; a full-body model that adds latency without improving outcomes is not automatically a better contribution.

10.10 The social-validity gap is contextual, not the absence of human research

SCAND includes human evaluation, Habitat 3.0 supports human-in-the-loop interaction, and the guidelines/proxemics work explicitly studies how social claims should be evaluated [17, 24, 34, 37]. It would therefore be incorrect to motivate the project by saying that existing social navigation has never considered people themselves.

The narrower question is whether behavior that improves the measured collision–completion tradeoff on a humanoid is also experienced as more predictable, comfortable and appropriate in a declared task. Morphology, sensor height, arm motion, carried objects and stopping style can all affect that relationship. An informative study would control speed and clearance while varying the response-aware decision mechanism, use counterbalanced conditions, and report both

objective outcomes and participant-level judgments. The results would then support a bounded statement about that task and population, not a universal preferred personal-space shape.

10.11 What this synthesis supports

Across the themes, the most defensible opening is an *auditable coupling* between observation age, uncertainty, executable humanoid response and the safety–liveness decision. Existing work establishes the individual ingredients and several combinations. The project must therefore demonstrate either a specific new mechanism with an advantage under matched conditions, or a new controlled explanation of failure mechanisms that changes how systems should be designed. The next section's gap priorities and the final novelty roadmap state the evidence needed to move from that opening to a contribution claim.

11 Comparative synthesis and ranked research gaps

Table 2: Closest families and the remaining project-specific question. This is a qualitative synthesis, not a benchmark ranking.

Prior work	Established capability	Question still testable here
DWA, VO, TEB, biped crowd navigation	Dynamics/feasibility matter in avoidance	Does a calibrated policy-level response envelope outperform generic bounds?
CADRL, SARL, RGL, DS-RNN	Learned spatial/temporal crowd reasoning	What happens under matched sensing age, misses, and G1 execution?
ASAP, RMA, perceptive locomotion	Adaptation and dynamics-aware learned control	Can frozen-policy response calibration improve navigation without retraining locomotion?
Safe humanoid policy, Omni-Perception	Sensor-conditioned legged/humanoid avoidance	How do modular filtering and end-to-end policies trade contact risk against liveness?
FLUX, OmniNavBench	Cross-embodiment navigation and richer evaluation	How much apparent performance changes with information provenance and measured execution delay?
Social-navigation guidelines and proxemics taxonomy	Multidimensional human-centered evaluation	Which safety/progress gains correspond to judged comfort and legibility?

11.1 Gap 1: calibrated executable avoidance under stale observations

Priority: highest; fit: direct; novelty confidence: moderate and conditional. Established feasible-velocity methods, biped planning, and recent LiDAR policies already address parts of the problem [6, 15, 16, 48]. The narrower unresolved question for this platform is the contribution of a *measured policy-level response distribution* combined with observation-age-aware person occupancy.

Hypothesis H1. At matched clearance conservatism and compute budget, a calibrated response envelope lowers contact incidence or increases collision-free mission completion compared with instantaneous and hand-tuned response models.

Contribution required. Define what is calibrated: delay, gain, acceleration/braking asymmetry, yaw response, combined-axis coupling, and residual prediction error. Condition or strat-

ify by relevant initial motion. Use held-out commands and scenes, and propagate residual uncertainty into the navigation decision. The estimator and its uncertainty treatment must be specified sufficiently to reproduce.

Falsification. If an equally conservative fixed-margin baseline matches the result, or improvements disappear on held-out response tests, the extra model complexity is not justified. A finding that timing dominates model complexity would still be useful if established through controlled interventions.

11.2 Gap 2: causal explanation of the oracle-to-onboard gap

Priority: high; fit: direct; novelty confidence: moderate. Benchmarks and perception datasets already exist [5, 32, 43]. What can be distinctive is a source-audited experimental decomposition on the *same walking system*.

Hypothesis H2. During close encounters, track age and partial-body velocity bias explain more avoidable failures than map or localization error alone, after controlling for scenario and locomotion policy. This is a hypothesis suggested by the repository's diagnostics, not an established causal result.

Contribution required. Independently replace observed person tracks, estimated robot pose, and online map with their oracle counterparts. Keep goal definitions, actor motion, policy, and evaluation truth fixed wherever possible. Then inject controlled delay, track dropout, identity changes, and pose drift. Log which information was actually consumed at the source selection branch, rather than relying only on launch arguments.

Falsification. If failures persist under oracle people but disappear under a changed low-level policy, the principal cause is likely execution rather than perception. If the accepted goals or actor exposure change between sensing modes, the comparison is confounded and needs redesign.

11.3 Gap 3: coordinated safety and liveness across planning layers

Priority: high; fit: direct; novelty confidence: moderate-to-low until a specific decision rule is defined. Freezing, cooperation, and hierarchical navigation are already studied [11, 22, 27, 45]. The opening is not the existence of a global/local hierarchy, but a verified policy for resolving disagreement between route intent, social planning, and safety supervision.

Hypothesis H3. Response-consistent handoff and event-driven route updates reduce prolonged veto loops and timeout probability without increasing contact or fall rates compared with fixed-rate replanning and independent supervision.

Contribution required. Define the state machine or optimization criterion governing yield, resume, detour, and route replacement. Measure delivered-route age, discarded or stale routes, supervisor disagreement, recovery time, and progress after a yield. Match total CPU budget and prediction assumptions across alternatives.

Falsification. Higher route-update frequency without improved collision-free completion is an engineering change, not confirmation of the hypothesis. Likewise, reduced timeouts bought by more contacts do not establish a better trade-off unless the full frontier is reported.

11.4 Gap 4: full-body risk beyond planar clearance

Priority: medium; fit: requires extensions; novelty confidence: conditional. Whole-body control and three-dimensional avoidance already exist [13, 20, 47, 48]. A useful narrower question is when a planar swept footprint is inadequate for humanoid social navigation.

Hypothesis H4. A validated body-swept-volume approximation detects encounter risks missed by base clearance alone, especially during turning, lateral stepping, and recovery.

This requires link-level contact/clearance reference measurements and scenario designs that exercise torso/arm motion and stability. Start by quantifying the error of the current footprint abstraction. A more expensive full-body method is warranted only if that error affects decisions. Claims about contact injury or human safety would need a different validation scope than collision counting.

11.5 Gap 5: contextual comfort and nonreciprocal crowds

Priority: longer term; fit: requires human evaluation and richer actors. Social principles and contextual proxemics are well developed [17, 34]. The opportunity is to test whether response-aware motion creates behavior people actually find more predictable and comfortable, while avoiding an assumed universal radius.

Compare nonreactive, reciprocal, and data-driven pedestrian models first. Then evaluate selected robot behaviors with human judgments, controlling for speed, clearance, and morphology. Without this step, use the term *human-aware collision avoidance* or *proxemic-cost navigation* for the demonstrated capability rather than claiming broadly validated social intelligence.

12 A concrete experimental programme

12.1 Stage A: characterize the frozen locomotion interface

Collect command step, ramp, stop, turn, and combined-axis responses across initial speeds, directions, and repeated runs. Record requested command, normalized command, supervisor output, policy-consumed command, estimated state, evaluation state, and timestamps. Measure latency, gain, overshoot, stopping distance, lateral drift, and fall outcomes. Separate training, calibration, and held-out response sets by complete trajectories, not adjacent samples from one rollout.

Compare (i) instantaneous planar integration; (ii) current hand-set envelope; (iii) fitted deterministic response; and (iv) fitted response with residual uncertainty. Evaluate horizon-dependent error and interval coverage. Software slew limits are not automatically physical acceleration bounds. Calibrate the active profile and policy together; switching either changes the model.

12.2 Stage B: controlled oracle planning study

Freeze maps, source information, policy, robot geometry, actor trajectories, and goals. The repository's development methods—Nav2-only, reactive geometric, deterministic temporal, and full planner/supervisor—provide a starting matrix, but are not substitutes for published comparator implementations. Add a well-specified classical baseline, preferably DWA, VO/ORCA with explicit reciprocity settings, or TEB, and describe adaptation to the G1's command interface.

All methods must share unavoidable hardware/policy limits. If the same supervisor is applied to every method, state that the comparison concerns the upstream planner under a shared shield. If supervision itself is the independent variable, separate the experimental arms so that a shared filter does not erase the difference being measured. Match tuning budgets; select parameters on development scenarios only.

12.3 Stage C: factorial onboard robustness study

Use a small, interpretable factorial design before scaling to many methods. Suggested factors are response model, observation-age compensation, perception source, and forecast uncertainty treatment. Add localization/map substitutions as diagnostic blocks if the full factorial is too costly. Report interactions: for example, a response model may help only when track age is corrected.

Scenario families should include head-on passing, orthogonal crossing, overtaking, occlusion emergence, corridor blockage, narrow passage, moving crowds, and turn/stop transients. Vary crowd density, human-to-robot speed ratio, sensor dropout, map ambiguity, and actor reciprocity. Include people-free trials to isolate locomotion and route-following failures. Hold out at least one scene family to probe overfitting to the plaza.

Identical seeds are not sufficient for pairing live runs: wall-time scheduling and observation-dependent goal admission can change encounters. For causal replay, freeze actor trajectories and goals where feasible, record accepted events, and quantify residual nondeterminism. Treat nonreactive replay and interactive crowds as separate experimental regimes.

12.4 Stage D: metrics and statistical interpretation

Predeclare primary outcomes: complete mission without contact/fall, contact incidence, and algorithmic timeout. Secondary outcomes include minimum body clearance, contact count/severity proxy, time in specified proxemic regions, approach speed, stopping error, path/time efficiency, yield/recovery duration, supervisor interventions, and processing/observation/actuation latencies. State whether a contact metric is based on geometry, physics contacts, or a distance threshold.

Keep four denominators visible: planned trials, executed trials, valid trials, and successful trials. A collision is an algorithmic outcome; a missing required scoring stream is an experiment-validity problem. However, method-induced overload can itself cause missing data. Predeclaring failure classification and reporting invalidity by method prevents invalid-trial exclusions from hiding poor computational robustness.

Use confidence intervals for rates and paired scenario/seed differences where replay supports pairing. Bootstrap at the independent scenario/seed unit, with repetitions nested within it, not at individual telemetry samples. The repository's Wilson intervals and paired aggregation are useful starting points, but the statistical design still needs to specify the independent unit and missingness assumptions. For zero observed collisions in n independent trials, the one-sided exact 95% upper bound is

$$p_{\text{upper}} = 1 - 0.05^{1/n}. \tag{6}$$

Even zero events in 100 independent trials leaves an upper bound near 3%. Repeated correlated scenarios do not justify the same interpretation. Choose the trial count from desired precision or a prespecified detectable effect, not from the first sequence of passing demonstrations.

Do not rank methods only by travel time on successful trials: this conditions on success and can reward a fast but unreliable method. Report completion and failure distributions alongside conditional efficiency. Avoid a single weighted score obscuring collisions, falls, invalidity, and timeouts.

12.5 Minimum publishable evidence package

1. A frozen protocol, policy/configuration hashes, source manifests, scenario definitions, and explicit information boundaries.
2. Held-out response-model error and calibration results.

3. Matched baselines plus ablations isolating response, delay, uncertainty, and supervision.
4. Multi-scenario, multi-seed mission results with retained failures, validity accounting, and uncertainty estimates.
5. Failure-case analysis linking sensor observations, decisions, applied commands, and physical outcomes.
6. Reproducible artifacts and a bounded contribution statement supported by those experiments.

13 A novelty roadmap: what the work must become

13.1 Separate an engineering milestone from a research contribution

A functioning simulator, an integrated ROS graph, a passing unit test or an isolated delivery is an engineering milestone. It can be essential to research without being a new scientific result. Likewise, correcting a stale timestamp or a missing subscription is necessary experimental hygiene; it becomes a research contribution only when embedded in a generalizable method or a controlled finding about the effect of that mechanism.

The route to novelty should consequently be expressed as three questions:

1. **What does the nearest prior work already solve?** Name the closest method and its assumptions, rather than comparing only with a weak default.
2. **What precisely changes?** Specify the new estimator, uncertainty construction, decision rule, coordination contract or experimental intervention.
3. **What observation would demonstrate or refute its value?** Define held-out outcomes and a fair alternative that could plausibly match the result.

The following roadmap is proposed work. It does not report completed acceptance, guarantee a publication, or certify novelty throughout literatures outside the reviewed corpus.

13.2 Primary contribution target: a joint observation–execution model

Intended contribution. A measured, policy-conditioned short-horizon motion envelope combined with acquisition-age-aware pedestrian occupancy, used consistently to select and supervise commands on a walking humanoid.

The prior-art boundary is explicit. DWA already reasons about reachable and stoppable velocities; the biped MPC paper already combines balance, uncertainty and stopping; ASAP already learns a dynamics correction; and recent LiDAR policies already perform sensor-driven avoidance [6, 16, 20, 47, 48]. The proposed method would need to contribute the *particular measured interface and decision construction*, not the generic combination of their keywords.

One candidate formulation represents executable robot occupancy as

$$\mathcal{R}_\theta(\tau \mid x, u, c) = \widehat{\mathcal{R}}_\theta(\tau \mid x, u, c) \oplus \mathcal{E}_\theta(\tau, c), \tag{7}$$

where c identifies policy/configuration and relevant posture/load conditions, and $\mathcal{E}_\theta$ is a residual envelope evaluated on held-out response trajectories. A person's predicted occupancy is aligned from the observation's acquisition time, not reset by republishes. Candidate acceptance then concerns overlap between the two occupied regions over the executable horizon. This is a *research specification*, not an already proved guarantee; valid coverage, dependence assumptions and feasible fallback behavior must be established.

Goal N1.1: characterize the interface. Freeze the official or explicitly selected payload policy, gains, filtering and command bounds. Measure steps, ramps, stops, reversals, pivots and

combined-axis transitions from varied initial motion. Identify deadbands, asymmetric gains, delay, stopping drift and cross-axis effects. Split calibration and test data by complete trajectories and conditions rather than adjacent samples.

Goal N1.2: define a parsimonious model. Compare instantaneous dynamics, generic bounded acceleration, the current hand-tuned envelope and a measured model with and without calibrated residual uncertainty. Justify additional parameters by held-out prediction or decision value. A more complex model that only fits its training motions is not progress toward this contribution.

Goal N1.3: demonstrate a downstream advantage. Under matched sensing, policy, actor schedule, goals and computational budget, show either lower contact risk at similar completion or improved collision-free completion at comparable risk. Report the frontier across conservatism settings instead of choosing one favorable threshold. An equally conservative fixed-margin baseline is a required alternative, because its success would refute the need for the extra model.

13.3 Secondary contribution target: causal diagnosis of the onboard gap

Intended contribution. A controlled, source-audited decomposition of how sensing, time alignment and execution cause navigation failures on the same walking system. Benchmark platforms and perception datasets already exist [5, 32, 43, 49]; the contribution must be new causal findings or a validated protocol that makes previously confounded effects measurable.

Goal N2.1: define matched interventions. Independently substitute person observations, robot pose and map information while preserving evaluation truth. Hold goals and actor exposure fixed when testing an algorithmic change; otherwise a source mode may change which encounters occur. Add controlled age, dropout, center bias, association errors and pose drift, with identities that allow a consumed observation to be linked to a resulting action.

Goal N2.2: measure interactions. A small factorial design should test response modeling against age correction and uncertainty treatment. Do not assume that errors add independently. A result that tracking age dominates predictor complexity, or that locomotion response dominates map error, is potentially useful only if the experiment isolates that effect and survives held-out conditions.

Goal N2.3: publish explanatory failures. For representative contacts, timeouts and invalid trials, reconstruct observation, estimate, proposal, veto, policy-consumed command and measured response. Report the frequency of the identified mechanism, not only one attractive anecdote. If source substitutions change the accepted goals or crowd schedule, classify the experiment as confounded until it is redesigned.

13.4 Third contribution target: consistent supervision with liveness

Intended contribution. A decision or handoff rule that keeps route following, social planning and final command supervision consistent about the motion they are checking, and measurably reduces prolonged veto loops without increasing contact or fall risk.

Freezing, cooperation and hierarchical navigation already have substantial prior art [6, 11, 27, 45]. A new state machine with names such as yield, wait and recover is not enough. The contribution would need a defined invariant or optimization criterion, a bounded action-lifecycle contract, and evidence explaining why the rule changes the tradeoff.

Goal N3.1: compare against shared and independent shields. Keep the underlying predictor and response model fixed while varying coordination. If a common final filter is attached to every planner, explain that the experiment tests planners under a shared shield. If

the shield is the independent variable, do not let another hidden filter erase the difference.

Goal N3.2: measure actual liveness. Include repeated-veto duration, command reversals, time to resume, useful progress after yielding, route age, interventions and complete goals. A stationary robot can still be struck by an approaching person; it is not a universally safe fallback. A controller that avoids contact by remaining blocked forever also fails the intended task.

Goal N3.3: test adverse scheduling. Evaluate cancellation during inference, late results, sensor starvation, competing command sources and changes of mission generation. Some fixes here are baseline correctness rather than novelty, but omitting them can invalidate the experiments used to support a new coordination rule.

13.5 Extension target: body, arm and payload risk

Intended contribution. Establish when a base-only footprint/envelope is insufficient for a humanoid, and develop a sufficiently efficient approximation of articulated swept occupancy if the evidence demonstrates a need.

Whole-body motion and capability-aware navigation already exist [13, 20, 22, 27]. The first goal is therefore an *abstraction-validity study*: measure disagreement between the planner's base approximation and link/payload clearance or native contacts. Include turns, lateral motion, reach/carry transitions and recovery. Distinguish body contact, object contact, fall and drop from personal-space intrusion.

If the simpler approximation is adequate in the declared conditions, the result supports keeping it. If it misses important risks, compare a posture-conditioned envelope with full articulated checking at matched computational budgets. Demonstrating a useful fidelity–latency tradeoff would be a stronger contribution than replacing every check with an expensive model without measuring its effect. Magnetic fixture completion remains separate from physical finger grasping.

13.6 Extension target: contextual human validation

Intended contribution. Determine whether a response-aware humanoid motion strategy improves perceived predictability or comfort in a defined interaction, beyond changes explained by speed and clearance.

SCAND already reports human evaluation, and the guidelines and proxemics taxonomy provide substantial methodology [17, 24, 34]. The new question must therefore be humanoid/task-specific, not "add a questionnaire". Start with a bounded, counterbalanced study using matched encounter geometry. Separate morphology, payload, yielding style, motion smoothness and clearance where possible. Use participant-level analysis, declare the instrument and population, and assess objective task outcomes alongside subjective responses.

Simulation video or human-in-the-loop avatars can be an intermediate instrument, but their relationship to physical encounters must be stated [37]. A gain in geometric clearance alone cannot close this research goal. Equally, a favorable perception rating cannot replace contact and stability evidence.

13.7 A staged evidence ladder for the first paper

Gate	Deliverable	Evidence required before the claim advances
E0: honest baseline	Frozen source/policy/configuration and functioning scorer	Freshness, source and contact definitions verified; failed and invalid trials retained.
E1: measured model	Held-out command-response dataset	Prediction error and coverage by horizon, initial motion and operating condition; calibration kept separate from test.
E2: specific method	Reproducible model/decision specification	Equations, uncertainty assumptions, fallback behavior and timing budget; nearest methods identified.
E3: attribution	Matched baselines and ablations	Equal information and unavoidable execution limits; generic bounds and fixed margins can challenge the proposal.
E4: generalization	Multi-scenario and held-out evaluation	Complete-mission outcomes with denominators, uncertainty, repeated runs and evidence against scene-specific tuning.
E5: explanation	Failure taxonomy and trace examples	Mechanism frequency and representative counterexamples, not only selected successes.
E6: bounded claim	Manuscript and reproducibility package	A claim limited to demonstrated conditions, with unresolved access/literature gaps and negative results disclosed.

This ladder is a research-evidence sequence, not a replacement for the engineering S0–S7 project plan. Code availability, a clean build and regression coverage support E0. They do not automatically satisfy E1–E6. Likewise, an all-pass development matrix is a useful milestone but does not by itself demonstrate novelty or statistically strong generalization.

13.8 What would make the first submission defensible?

The strongest near-term submission should have one central contribution rather than claim every gap at once. A sensible scope is N1 supported by the diagnostic tools of N2, with N3 included only if a well-defined coordination mechanism is actually necessary to realize the model's benefit. Full-body and human-evaluation extensions can then form later studies with their own evidence requirements.

Before submission, extend the nearest-prior-work search specifically for uncertainty-aware biped MPC, delayed/sampled-data safety filters, learned reachable sets, decision-focused prediction and uncertainty calibration. Compare the proposed mathematical object, not just method names. The four inaccessible originals in this review should also be obtained where possible. A claim of first-ever capability is unnecessary and fragile; a clearly demonstrated conditional improvement or explanatory negative result is more defensible.

The intended result should ultimately be expressible in one sentence with a declared comparator and scope: *under matched sensing, controller and crowd conditions, the proposed measured observation–execution model improves the collision-free completion tradeoff, and controlled ablations identify when that improvement depends on age, response calibration or uncertainty.* If the data do not support that sentence, report which mechanism failed and revise the research

claim rather than loosening the evaluation definition.

14 Recommended positioning and conclusion

Suggested working title: *Observation-Age-Aware, Response-Calibrated Social Navigation for a Walking Humanoid: A Source-Audited Evaluation in MuJoCo and ROS 2.*

This title describes a proposed research direction, not completed results. The paper's eventual central claim should take a falsifiable form:

> Under a declared sensing, locomotion, and crowd model, the proposed calibration and decision procedure improves collision-free mission completion relative to matched baselines, and controlled ablations identify when the improvement depends on observation age and executable response.

Whether that statement becomes true is an experimental question. At present, the repository supports the platform and diagnosis portions of the work, while reliable onboard avoidance remains open.

The literature gives a clear boundary to novelty. Dynamics-aware avoidance, biped crowd planning, learned crowd interaction, sensor-driven G1 avoidance, cross-embodiment navigation, and physically grounded navigation benchmarks already exist. The most promising contribution is a carefully controlled account of their integration failure modes, together with a specific, validated mechanism that improves the safety–liveness trade-off. A rigorous negative result showing that a proposed response model does not help under realistic sensing would be more informative than an unqualified success claim based on selected oracle runs.

A Annotated evidence map of the 50 retained works

The entries below give each work's role in this review. They are not claims that every work received an equally detailed full-text review. Identity records and content-inspection levels are listed separately in the audit.

Work	Contribution to the synthesis and transfer limitation
Kruse et al. [25]	Foundational human-aware navigation framing; supplies historical context rather than a contemporary G1 baseline.
Singamaneni et al. [42]	Faceted taxonomy of socially aware navigation; guards against reducing social behavior to clearance alone.
Learning-based review [2]	Recent comparative overview and benchmark perspective; motivates learned baselines without transferring its rankings to different sensing contracts.
Evaluation guidelines [17]	Defines multiple social-navigation principles and evaluation dimensions; supports separate safety and human-centered outcomes.
Nahum et al. [34]	Context-dependent proxemics taxonomy; argues against universal personal-space geometry as a complete comfort metric.
MuJoCo [44]	Physics-engine foundation; simulator choice alone is neither calibrated dynamics nor algorithmic novelty.
ROS 2 [31]	Middleware architecture and deployment context; application-level latency remains a measured property.
The Marathon 2 [30]	Nav2 architecture and navigation orchestration; not a G1 policy-response model.

Work	Contribution to the synthesis and transfer limitation
SLAM Toolbox [29]	Mapping infrastructure; online mapping success must be separated from crowd avoidance.
DWA [16]	Dynamics-constrained velocity search baseline; makes generic acceleration-aware novelty claims inappropriate.
Velocity obstacles [15]	Relative-motion collision reasoning and admissible velocities; requires appropriate motion estimates and robot constraints.
ORCA [46]	Reciprocal collision-avoidance reference; actor reciprocity must match or be explicitly varied.
Sparse trajectory optimization [38]	Time/trajectory optimization comparator; the deployed controller model remains consequential.
Social force [21]	Interpretable pedestrian interaction model; not universal empirical human behavior.
Trautman et al. [45]	Dense-crowd cooperation perspective; motivates explicit freezing/progress metrics.
CADRL [12]	Learned decentralized collision avoidance; representation and training crowd assumptions need matching.
SARL [10]	Attention-based crowd interaction reasoning; useful learned-state baseline, separate from person detection.
RGL [11]	Graph relations and predictive planning; compare under the same sensing and execution model.
DS-RNN [28]	Temporal/spatial learning and partial observability; real wheeled transfer does not establish biped response equivalence.
SocNavBench [5]	Grounded, common evaluation scenarios and metrics; already occupies broad benchmark-novelty territory.
SocialGym 2.0 [9]	Configurable multi-agent navigation environments; robot interaction assumptions affect transfer.
JRDB [32]	Egocentric human perception dataset; relevant to misses/association, not a complete G1 closed-loop benchmark.
SCAND [24]	Human navigation demonstrations across robot morphologies; behavioral reference with different sensors and actuation.
DROW [4]	Learned sparse-range detection precedent; wheelchair-oriented original task differs from torso-mounted G1 person tracking.
DR-SPAAM [23]	Temporal/spatial person detection from range scans; projection and walking-motion domain shift require evaluation.
FAST-LIO2 [50]	Strong direct LiDAR–inertial estimation comparator; localization improvements alone do not establish navigation reliability.
ORB-SLAM3 [8]	Visual/visual–inertial state-estimation alternative; requires a suitable camera pipeline.
Social LSTM [1]	Interaction-aware learned trajectory prediction; forecasting and decision quality are different outcomes.
Social GAN [19]	Diverse plausible futures; sample diversity does not by itself calibrate risk.
Trajectron++ [41]	Dynamical, heterogeneous, optionally ego-conditioned prediction; an important alternative to independent constant velocity.
Rudenko et al. [39]	Forecasting taxonomy, context, datasets, and metrics; motivates matching the prediction task to planning.
CBF-QP [3]	Formal safety filtering under assumptions; distinguishes invariant conditions from empirical rollout checks.
Safe learning survey [7]	Separates safe control, robustness, and certification; prevents overclaiming trained or heuristic safety.
Biped crowd navigation [6]	Direct prior art combining viability, uncertainty and stopping; the expansion inspects its MPC hierarchies and numerical failure analysis.
RMA [26]	Rapid locomotion adaptation; adaptation competence is not a pedestrian-planning policy.
Massively parallel walking [40]	Efficient locomotion training; does not establish social mission completion.

Work	Contribution to the synthesis and transfer limitation
Perceptive locomotion [33]	Robust exteroceptive/proprioceptive integration; primarily terrain competence rather than social acceptability.
ANYmal Parkour [22]	Capability-aware navigation hierarchy; demonstrates that embodiment-aware planning already exists.
Wheeled-legged autonomy [27]	Integrated mobility-aware navigation and long-range missions; different morphology but strong systems-level prior art.
ExBody [13]	Expressive humanoid control with locomotion; perceived comfort needs an additional evaluation.
DeepMimic [35]	Physics-based motion imitation foundation; animation/skill objectives differ from crowd safety.
Dynamics randomization [36]	Foundational sim-to-real robustness method; not a calibrated stopping certificate.
ASAP [20]	Residual action alignment and G1 transfer; full-text methods/tables examined, task is motion imitation rather than social navigation.
Safe humanoid locomotion [48]	Closest LiDAR/G1 safety/comfort competitor; inspected costs and tables support a qualified, scenario-dependent comparison.
FLUX [18]	Recent generative navigation and G1 transfer; inspected metric definitions and deployment timing constrain cross-paper comparisons.
OmniNavBench [43]	Physically embodied, multi-task benchmark; inspected protocol narrows the open benchmark contribution.
MetaUrban [49]	Compositional urban micromobility simulation; inspected task/metric definitions expose the cost–completion tradeoff.
Habitat 3.0 [37]	Human/avatar collaboration and human-in-loop evaluation; inspected sensor contracts differ from an unprivileged G1 pipeline.
DPed-VLN [14]	September 2026 dynamic-pedestrian language-navigation benchmark; inspected tables separate goal success from human-related risk.
Omni-Perception [47]	Raw-LiDAR legged avoidance; inspected architecture, ablations and limitations require careful interpretation of success and collision metrics.

B Detailed paper-by-paper critical appraisals

The following structured records are part of the review, not a substitute for its thematic synthesis. Each separates a method's contribution, the inspected evidence, critical interpretation and implications for the project. "Selected full text" means the identified passages were inspected; it does not mean code replication, verification of every appendix, or equivalence between a preprint and its later version of record. Record-limited entries remain explicit evidence limitations.

B.1 Foundations and social requirements

B.1.1 From obstacle avoidance to human-aware navigation

Publication: Kruse et al. [25]. *Human-aware robot navigation: A survey.* Publication record.

Evidence anchor: Selected full-text inspection. Author manuscript: introduction, human-awareness requirements, Section 3 architecture discussion. Inspected source copy.

Method and contribution. Kruse and colleagues organize navigation around requirements arising from people, rather than treating a person as another occupied cell. Their discussion

connects comfort, socially appropriate motion and task-specific interaction to perception, prediction, local planning and deliberate behavior. The architecture distinguishes reactive responses from decisions such as following, guiding, approaching and resolving a blocked passage. This is an important conceptual foundation: social navigation is a coupled perception-and-behavior problem, not a particular local planner.

What the evidence establishes. The evidence is a literature synthesis and architectural classification. The inspected discussion lists concrete behaviors such as reducing speed near a person, avoiding erratic motion and approaching appropriately for an intended interaction. It does not provide a common experimental rerun of all surveyed algorithms.

Critical appraisal. The breadth is useful for defining requirements, but a taxonomy is not a validated universal utility function. Different behaviors can conflict: a direct approach may be appropriate for interaction and intrusive during independent passage. Its age also means that contemporary end-to-end humanoid results must be assessed separately. The review should not be used to characterize the capabilities of all later learning methods.

Implication for novel work. Use this paper to define the task before selecting metrics. GeoffSim currently addresses independent navigation and experimental delivery; it should not claim guiding, negotiation or perceived comfort merely because it avoids a moving actor.

B.1.2 A multifaceted taxonomy of socially aware navigation

Publication: Singamaneni et al. [42]. *A survey on socially aware robot navigation: Taxonomy and future challenges*. Publication record.

Evidence anchor: Selected full-text inspection. Accepted manuscript: abstract and taxonomy definitions, especially planning/decision-making, situation awareness and evaluation diagrams. Inspected source copy.

Method and contribution. Singamaneni and colleagues replace a single algorithm hierarchy with several complementary taxonomies. The inspected definitions distinguish implicit and explicit communication, independent/assistive/collaborative tasks, negotiation, local motion generation and global decision-making. Situation awareness includes representing agents, predicting their evolution and interpreting environmental context. Evaluation is treated as a separate dimension rather than inferred from the planner's architecture.

What the evidence establishes. The review maps a broad preceding decade of work and identifies relationships among representations, algorithms, tasks and evaluation tools. Its principal output is a structured description of the field, not a numerical leaderboard. The faceted view permits two systems using different mathematical methods to be compared on the same task or information requirement.

Critical appraisal. Taxonomic coverage does not establish that any combination of components delivers reliable autonomy. Labels such as learning-based or social remain too coarse to determine deployment privilege, latency or physical contact fidelity. A framework may cover several categories while still omitting the interactions between them. This makes the taxonomy a design and reporting instrument, not a substitute for source-level comparison.

Implication for novel work. Organize Geoff-Sim's contribution across task, information, embodiment and evidence dimensions. Its most promising novelty lies at an interface between those dimensions, not in introducing another instance of an already populated algorithm family.

B.1.3 Learning-based social navigation and the role of input contracts

Publication: Alyassi et al. [2]. *Social robot navigation: a review and benchmarking of learning-based methods.* Publication record.

Evidence anchor: Selected full-text inspection. 2025 publisher PDF: Tables 1, 2 and 4; Sections 2.4 and 2.5 on prediction, hybrid planning and safety-aware navigation. The stable citation key does not denote its publication year. Inspected source copy.

Method and contribution. Alyassi and colleagues classify learning methods by what information they consume and how they make decisions. The tables separate raw-sensor end-to-end methods, human-state methods, attention-based interaction models and prediction-based planners. The discussion covers hybrid combinations in which learned subgoals or pedestrian forecasts are passed to model-predictive control, as well as policy switching and fallback mechanisms.

What the evidence establishes. The inspected tables explicitly record architecture, training environment, output interface, code availability and real-world demonstrations. The prediction discussion reports that sophisticated forecasts do not uniformly dominate constant-velocity alternatives in every downstream application. Its benchmarking context is useful, but this appraisal does not extract a universal ranking or pool its outcomes with Geoff-Sim trials.

Critical appraisal. The strongest lesson is that architectural labels conceal different information contracts. A detector-based method and a state-vector policy solve different parts of the pipeline. Secondary descriptions of a method are useful for discovery but do not replace reading its primary experiment. Hybrid planning, uncertainty handling and fallback switching are already established directions; combining them generically is not itself a novelty claim.

Implication for novel work. Use this review to select matched baselines and identify additional primary literature before submission, particularly robust MPC and uncertainty-aware hybrid policies. Treat proposed gaps in this document as conditional on the bounded primary corpus.

B.1.4 Social-navigation principles and evaluation design

Publication: Francis et al. [17]. *Principles and Guidelines for Evaluating Social Robot Navigation Algorithms.* Publication record.

Evidence anchor: Selected full-text inspection. HAL manuscript: principles/guidelines overview and scenario-design methodology, including scenario cards. Inspected source copy.

Method and contribution. Francis and colleagues distinguish safety, comfort, legibility, politeness, social norms/competency, understanding of agents, proactivity and contextual appropriateness. They connect these principles to experimental design, metrics, scenarios, simulators, benchmarks and datasets. Scenario cards specify research context, intended robot task, intended human behavior and success criteria, making the experimental question explicit before a metric is calculated.

What the evidence establishes. This is a community methodological synthesis rather than a new robot controller. The inspected scenario-design section explains why scenarios should be representative and flexible, instead of prescribing the intended solution so narrowly that an algorithm can succeed without general competence. It also separates the quality of an evaluation instrument from the quantity of measurements collected.

Critical appraisal. A principle list does not supply a single scalar social score. Some dimensions require human judgments or a context-specific instrument; others can be measured geometrically. Counting contacts does not measure politeness, and rating a short video does not establish physical reliability. The accessible manuscript predates the final bibliographic record, so version-specific wording must remain identifiable.

Implication for novel work. Geoff-Sim should publish scenario cards, source contracts, failure categories and separate outcome families. This is a methodological baseline to adopt, not a new contribution to claim. Novelty must come from the scientific question or new validated mechanism tested with it.

B.1.5 Proxemics as a contextual variable rather than a universal radius

Publication: Nahum et al. [34]. *Advancing a taxonomy of proxemics for socially aware robot navigation.* Publication record.

Evidence anchor: Selected full-text inspection. Publisher PDF: Section 3 methodology, Table 1 shape-model comparison and variable-classification discussion. Inspected source copy.

Method and contribution. Nahum and colleagues structure proxemic influences into human, robot, environment and context factors, alongside spatial outcome variables. Their method combines a top-down taxonomy with bottom-up extraction from experimental studies. The inspected table contrasts concentric circles, asymmetric and motion-dependent shapes, and emerging three-dimensional representations. Personal space is therefore modeled as contingent on the interaction, not as a fixed property of an isolated person.

What the evidence establishes. The paper describes a 2020-2025 search yielding more than 190 initial articles, 99 screened articles and 43 retained items, comprising four surveys and 39 experimental studies. These are the source paper's counts, not the screening counts of this review. Variables were manually coded, with generative-AI assistance for clustering, and underrepresented combinations were identified through frequency analysis.

Critical appraisal. An infrequently studied factor is not automatically a practically important research gap. The taxonomy and corpus-selection process determine which absences become visible. Context-specific findings also resist direct numerical aggregation into one clearance threshold. A robot's body size, speed, purpose and approach direction may change judgments even when center distance is identical.

Implication for novel work. Keep a fixed proxemic threshold as an explicitly declared engineering metric, then test contextual alternatives. A future humanoid-comfort contribution needs a study connecting motion, morphology and participant judgments, rather than relabeling a distance penalty as validated comfort.

B.2 Geometric avoidance, interaction and safety

B.2.1 DWA and the distinction between reachable and stoppable commands

Publication: Fox et al. [16]. *The dynamic window approach to collision avoidance*. Publication record.

Evidence anchor: Selected full-text inspection. Author technical manuscript: abstract, Sections 1-4, motion-model and dynamic-window explanation. Inspected source copy.

Method and contribution. Fox, Burgard and Thrun search directly in translational/rotational velocity space. The admissible set contains commands whose trajectories allow stopping before an obstacle, while the dynamic window restricts candidates to velocities reachable within the next control interval. The objective balances goal alignment, speed and obstacle clearance. Piecewise circular trajectories simplify collision checking for the synchro-drive model.

What the evidence establishes. The manuscript describes implementation on RHINO and other mobile robots, with reported operation up to 95 cm/s in populated environments. This is historical hardware evidence under the paper's platform and sensing conditions. It is not a contemporary multi-seed humanoid crowd benchmark, and its demonstrations do not define the exposure denominator needed for a reliability estimate today.

Critical appraisal. DWA directly contradicts a claim that classical avoidance ignores dynamics or braking. Its model is nevertheless a specific abstraction: planar velocity control and local trajectories do not automatically capture a learned biped policy's deadbands, asymmetry, balance transients or upper-body sweep. Local optimality and incomplete environmental information also leave global liveness problems unresolved.

Implication for novel work. The relevant extension is calibrated policy-level reachability under sensing age, compared against a properly tuned admissible-velocity baseline. Adding an acceleration cap to an otherwise instantaneous planner is an engineering correction, not sufficient methodological novelty.

B.2.2 Velocity obstacles as a relative-motion foundation

Publication: Fiorini and Shiller [15]. *Motion Planning in Dynamic Environments Using Velocity Obstacles*. Publication record.

Evidence anchor: Record-limited inspection. Verified DOI metadata and recorded abstract; attempted institutional full-text locator was unavailable.

Method and contribution. The velocity-obstacle formulation represents relative velocities that lead to collision with a moving obstacle and reasons about selecting motion outside that region. The retained record places dynamic-environment motion planning and velocity-space reasoning at the foundation of later reciprocal methods. The original work also discusses admissibility; the conceptual account here is deliberately separated from an uninspected reconstruction of its implementation.

What the evidence establishes. The primary bibliographic identity is verified, but the full paper was not retrieved in this expansion. Consequently, no new table values, hardware protocol,

runtime number or completeness claim is attributed to this publication. Detailed executable-model comparisons should use the accessible ORCA and DWA texts and return to this original article when access is obtained.

Critical appraisal. The conceptual strength is interpretability: a rejection can be related to relative motion and geometry. Transfer depends on the validity of state estimates, prediction assumptions and command realizability. Those are comparison questions, not evidence that the original paper neglects every such issue. Missing access must not be converted into a negative assessment of its methods.

Implication for novel work. Use VO as a foundational baseline family and novelty boundary. Resolve the original full text before making a first-of-kind claim about velocity admissibility, dynamic obstacles or uncertainty treatment.

B.2.3 ORCA: reciprocal guarantees and their execution assumptions

Publication: van den Berg et al. [46]. *Reciprocal n-Body Collision Avoidance.* Publication record.

Evidence anchor: Selected full-text inspection. Author chapter: introduction, reciprocal formulation, Section 6 experiments and Section 7 limitations. Inspected source copy.

Method and contribution. ORCA derives pairwise sufficient conditions for reciprocal collision avoidance, assigns each cooperating agent part of the required velocity correction and intersects the permitted half-planes. A low-dimensional linear program chooses a velocity close to the preferred one. The method separates local collision avoidance from the global route that provides that preference and is computationally attractive for large numbers of independently deciding agents.

What the evidence establishes. The experiments include circle exchange and office evacuation with thousands of simulated agents. Reported timings distinguish scenarios and hardware: for example, 5,000 agents required 8 ms in the circle case and 15.6 ms in the office case on eight cores. These are simulator-wide algorithm timings, not observation-to-actuation latency on a walking robot.

Critical appraisal. The conclusion explicitly states that its simple robot model ignores kinematic and dynamic constraints and that clamping a computed velocity afterward does not preserve strict collision avoidance. Reciprocity is equally important: nonreactive people do not implement their assigned correction. Thus performance under unilateral avoidance cannot be inferred from the reciprocal guarantee, and a baseline should disclose responsibility weights and post-processing.

Implication for novel work. Compare reciprocal and nonreciprocal actor regimes separately. Adaptation to the G1 must model what commands can actually be executed, while preserving a clearly documented baseline rather than silently changing ORCA's assumptions.

B.2.4 Sparse timed-trajectory optimization

Publication: Rosmann et al. [38]. *Efficient trajectory optimization using a sparse model.* Publication record.

Evidence anchor: Record-limited inspection. Verified DOI record and the authors' maintained teb_local_planner description; the original 2013 paper PDF was not retrieved.

Method and contribution. The timed-elastic-band family jointly adjusts local trajectory geometry and timing to trade execution time, obstacle separation and kinodynamic constraints. The authors' software description identifies this optimization structure and cites the 2013 sparse-model paper. This review uses that implementation-level description as context and does not assume that every feature of the later software was already present in the cited conference paper.

What the evidence establishes. The paper identity and its place in the method lineage are established. The current software repository is accessible, but source inspection and a successful software build are not experimental replication of the 2013 results. No uninspected original runtime, success rate or dynamic-obstacle guarantee is reported here.

Critical appraisal. An optimized trajectory is only useful if the downstream controller follows its timing and geometry. The important comparative issues are initialization, obstacle forecasts, constraint handling and execution mismatch. These are reasons to include timed optimization as a serious baseline, not proof that a simpler response-envelope method is superior. Later homotopy and topology extensions must be cited separately if evaluated.

Implication for novel work. Use an explicitly versioned TEB implementation for comparison, state its command model and shared supervisor, and obtain the original primary text before assigning paper-specific limitations or claiming a new time-aware planning formulation.

B.2.5 Social forces and emergent pedestrian organization

Publication: Helbing and Molnár [21]. *Social force model for pedestrian dynamics.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: abstract, force-model exposition and collective-behavior illustrations, including lane formation. Inspected source copy.

Method and contribution. Helbing and Molnar describe pedestrian movement using an acceleration toward desired velocity, repulsive interactions with people and boundaries, and possible attractions. The forces represent motivations rather than literal mechanical contact forces. Coupled stochastic equations produce collective effects such as directional lanes, showing how locally specified interactions can generate structured crowd motion without centrally planning every pedestrian.

What the evidence establishes. The inspected illustrations demonstrate emergent crowd organization and compare the modeled effects with observed pedestrian phenomena. This is evidence about a behavioral modeling mechanism, not a robot's ability to recognize social context or a universal empirical model of all people. The accessible arXiv deposit is later than the original journal publication.

Critical appraisal. Parameters, boundary conditions and desired directions strongly influence the resulting crowd. Using the same social-force model both to generate training data and to evaluate a learned robot can reward adaptation to the simulator's assumptions. Apparent smoothness or lane formation does not establish realistic responses to a humanoid, vulnerable road users, distraction or cultural conventions.

Implication for novel work. Treat crowd-generation policy as an experimental factor. GeoffSim's independent actors answer a different question from mutually adapting agents, and results should be repeated under more than one behavior model before claiming broad social competence.

B.2.6 Cooperation and the freezing-robot problem

Publication: Trautman et al. [45]. *Robot navigation in dense human crowds: the case for cooperation.* Publication record.

Evidence anchor: Record-limited inspection. Verified DOI record and Caltech institutional abstract; the landing record was metadata-only.

Method and contribution. Trautman and colleagues frame dense-crowd navigation as joint behavior rather than avoidance of independently predicted obstacles. The institutional abstract describes an extension of interacting Gaussian processes to multiple goals and stochastic movement duration. Cooperation is intended to reduce the excessive conservatism that arises when the robot considers other agents' possible futures without accounting for mutual adaptation.

What the evidence establishes. The accessible institutional abstract reports 488 completed runs and comparisons with teleoperation and noncooperative/reactive planners. These are abstract-reported results, not independently inspected protocols. This appraisal therefore avoids reconstructing participant allocation, unsafe-event adjudication or uncertainty intervals from that count.

Critical appraisal. The central insight is relevant even when the full protocol is unavailable: treating all possible occupancy as fixed exclusion can eliminate every route, while assuming cooperation can fail when people do not yield. A model of interaction must be evaluated in conditions where its behavioral assumptions are both satisfied and violated. Access limitations prevent a detailed original-table audit here.

Implication for novel work. Measure liveness explicitly: time blocked, progress after yielding and interventions, alongside contacts. A new G1 supervisor must not be presented as the first treatment of freezing or cooperative navigation; its novelty would need a specific execution- and observation-aware mechanism.

B.2.7 Biped viability, uncertainty and relaxed passive safety

Publication: Bohorquez et al. [6]. *Safe navigation strategies for a biped robot walking in a crowd.* Publication record.

Evidence anchor: Selected full-text inspection. HAL manuscript: Sections II-V; equations 4-12; hierarchy definitions; Tables I-V and failure discussion. Inspected source copy.

Method and contribution. Bohorquez and colleagues combine balance and collision avoidance using a linearized center-of-mass/center-of-pressure model, foot-placement constraints and a terminal capture/stopping condition. Lexicographic MPC gives physical feasibility and near-term collision avoidance priority over velocity tracking. Pedestrians move at constant velocity and do not avoid the robot; position and velocity uncertainty expand their occupied region over the horizon. Alternative hierarchies defer or relax the terminal stop requirement when strict passive safety becomes counterproductive.

What the evidence establishes. The numerical study uses an HRP-2-like model, 100 stored crowd realizations per condition, varying crowd size/speed, field of view, horizon and uncertainty. It explicitly excludes perception occlusions. The analysis relates observable range to closing speed and horizon, and distinguishes robot feasibility failures from collisions. Relaxing physical feasibility can reduce collisions by allowing the robot to fall, which is not an overall improvement.

Critical appraisal. This is unusually close prior art: dynamics, uncertainty, balance, stopping and nonreactive crowds are already integrated. Its simplified walking and perception models differ from a learned G1 controller with measured latency, but those differences must become a defined new modeling or decision contribution. Passive safety also does not prevent a person from subsequently colliding with a stopped robot.

Implication for novel work. This paper is a mandatory comparator and sharply narrows novelty. The defensible question is whether empirically calibrated policy response and acquisition-age uncertainty improve that safety-liveness tradeoff under actual onboard sensing, rather than whether those concepts should be combined at all.

B.2.8 Barrier-function guarantees are conditional statements

Publication: Ames et al. [3]. *Control Barrier Function Based Quadratic Programs for Safety Critical Systems*. Publication record.

Evidence anchor: Selected full-text inspection. Author manuscript: abstract; Sections II-IV definitions and invariance discussion; automotive application framing. Inspected source copy.

Method and contribution. Ames and colleagues formalize safety through forward invariance of a set and performance through Lyapunov-style objectives. Reciprocal and zeroing barrier constructions lead to admissible control conditions that can be enforced by a quadratic program. The optimization framework exposes conflicts between performance and safety instead of hiding both inside one undifferentiated reward. The mathematical guarantee belongs to the modeled system and stated regularity/feasibility conditions.

What the evidence establishes. The paper develops theoretical results and illustrates the framework with adaptive-cruise-control and lane-keeping problems. Its proof structure is different from an empirical collision-rate comparison. Neither automotive examples nor an inequality written in a learning objective establish a humanoid implementation's model accuracy, sensing coverage or sampled-data behavior.

Critical appraisal. A practical filter must address control limits, model error, observation uncertainty and whether a feasible safe action exists. A planner can produce a command that satisfies a nominal constraint and still fail if a later slew limiter or policy changes its realized trajectory. Replacing an online constraint with a finite-weight learning cost changes the assurance level and requires a separate argument.

Implication for novel work. Use barrier literature to define precise guarantee language and to design a model-matched shield comparator. Geoff-Sim's current rollout supervisor should be described by its measured rejection behavior and assumptions unless a valid invariance argument is actually established.

B.2.9 Learning robustness versus certifying safety

Publication: Brunke et al. [7]. *Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning.* Publication record.

Evidence anchor: Selected full-text inspection. Author survey: introduction and Section 3 taxonomy of learned dynamics, safe RL and certification. Inspected source copy.

Method and contribution. Brunke and colleagues organize safe learning around prior model knowledge, uncertainty and the type of guarantee sought. The inspected taxonomy separates learning uncertain dynamics within a control framework, encouraging safe behavior through reinforcement learning, and certifying or modifying a learned controller through constraints or backup mechanisms. This distinguishes a performance-improving learned model from a validated safety envelope.

What the evidence establishes. The work synthesizes control and learning methods, their assumptions and benchmark practices. It explains why guarantees depend on the formalized uncertainty and operating conditions. It does not certify an arbitrary controller merely because that controller uses a safety-related cost, a Bayesian model or a backup stop.

Critical appraisal. The taxonomy highlights an important reporting failure: robustness demonstrated on sampled disturbances is often described with the language of formal safety. Conversely, formal guarantees can become operationally weak when their assumptions exclude the failures that matter. There is also a difference between keeping learning exploration safe and keeping a fixed deployed policy safe after training.

Implication for novel work. Position a response-calibrated G1 method at the assurance level its evidence supports. Report model residual coverage, operating conditions and failure modes, then evaluate whether a stronger certified formulation remains feasible at the available compute and sensing budget.

B.3 Learning interaction and predicting motion

B.3.1 CADRL: amortizing interaction reasoning into a value function

Publication: Chen et al. [12]. *Decentralized non-communicating multiagent collision avoidance with deep reinforcement learning.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: abstract, robot-centric parameterization, value-function action selection and training algorithm. Inspected source copy.

Method and contribution. CADRL learns an estimate of progress-to-goal from the joint observable state of interacting agents. At runtime, candidate velocity actions are scored through short lookahead and the learned value function. A robot-centric coordinate transformation removes translation/rotation redundancy. Neighbor goals are hidden, and the implementation approximates their immediate evolution using filtered velocity. Multiagent use builds on a policy learned around simpler interactions rather than solving a full joint trajectory optimization at every step.

What the evidence establishes. The paper presents simulation comparisons with ORCA and reports improved travel-time quality under its protocol. The inspected formulation makes

the assumed reciprocity and state-transition approximation visible. Its main computational advantage is shifting expensive interaction learning offline, not eliminating uncertainty about other agents' intentions.

Critical appraisal. The observation interface already contains agent positions, velocities and radii, so detection, association and localization are separate responsibilities. Pairwise aggregation can miss collective effects, motivating later attention and graph methods. A learned state value is also not a physical feasibility certificate: candidate velocities still need a model of execution on the target robot. The meaning of the baseline comparison changes if a shared safety supervisor overrides most selected actions.

Implication for novel work. CADRL supplies an interpretable learned-state baseline. Compare it on matched tracks, action limits and supervision, and report any mismatch between its training crowd behavior and the nonreactive or independently moving actors used for evaluation.

B.3.2 SARL: learning which people matter collectively

Publication: Chen et al. [10]. *Crowd-Robot Interaction: Crowd-Aware Robot Navigation With Attention-Based Deep Reinforcement Learning.* Publication record.

Evidence anchor: Selected full-text inspection. arXiv v2 full HTML and PDF: Sections II-IV, implementation details and Tables I-II. Inspected source copy.

Method and contribution. SARL embeds robot-human interactions and pools them with learned attention; LM-SARL additionally represents human-human interactions with local maps. The weighted embedding gives a fixed-size crowd representation despite varying neighbor count. A value network selects among sampled velocity actions. The paper explicitly assumes holonomic motion and instantaneous achievement of the commanded velocity, which is a materially different interface from a delayed walking policy.

What the evidence establishes. The evaluation separates invisible robots, to which ORCA-controlled humans do not react, from visible robots with mutual influence, using 500 test cases in each setting. LM-SARL reports success 1.00 in the displayed tables; ORCA's invisible-setting success is 0.43. Real-world Segway demonstrations are also reported. These values describe a specific planar simulator and protocol, not complete sensing-to-motor reliability on a humanoid.

Critical appraisal. The paper's controlled ablations support attention and local interaction encoding within that task. They do not make attention weights causal explanations of human intent. Training with true next-state information and deployment with an approximate transition model require careful distinction, as emphasized by RGL's later comparison. The 0.2 m discomfort metric is a separation proxy rather than a human-rating instrument.

Implication for novel work. SARL is a strong architectural baseline for crowd aggregation, but its instantaneous-action assumption must be exposed in G1 comparisons. A contribution should show what calibrated execution and observation age add beyond better crowd-state encoding.

B.3.3 Relational graphs and multi-step lookahead

Publication: Chen et al. [11]. *Relational Graph Learning for Crowd Navigation.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: model/training description, Table I, planning-budget discussion and physical demonstration description. Inspected source copy.

Method and contribution. RGL learns interactions among agents through a graph convolutional representation shared by state prediction and value estimation. Predicting future states enables multi-step lookahead instead of relying solely on a one-step terminal value. The planning depth and retained action width create an explicit tradeoff between computation and anticipation. This is a learned-model planning approach, distinct from directly regressing the next command from sensor history.

What the evidence establishes. Table I reports five independently seeded training runs and 500 test cases for a five-person circle crossing. MP-RGL-Multistep reports success 0.96 plus/minus 0.02 and collision 0.02 plus/minus 0.01 under that protocol. The comparison replaces a ground-truth-next-state assumption with a linear prediction variant before assessing the learned relation/model contribution. Pioneer demonstrations provide additional qualitative deployment context.

Critical appraisal. The source-contract adjustment is central: architecture gains cannot be separated from privileged transition information otherwise. Additional lookahead can improve behavior but also magnify model errors and consume the time needed to react. An average planning time alone does not capture this feedback. The physical demonstrations do not provide a matched biped dynamics or latency study.

Implication for novel work. Use RGL to motivate joint reporting of prediction fidelity, planning horizon, computation and resulting mission outcomes. A G1 comparison should hold the observation contract and search budget fixed before attributing gains to response calibration or graph structure.

B.3.4 Temporal crowd reasoning under restricted visibility

Publication: Liu et al. [28]. *Decentralized Structural-RNN for Robot Crowd Navigation with Deep Reinforcement Learning.* Publication record.

Evidence anchor: Selected full-text inspection. Retrieved arXiv v4: experiment setup, field-of-view/group evaluation, real-world section and limitations. Inspected source copy.

Method and contribution. DS-RNN represents spatial and temporal relationships through structured recurrent components and trains a model-free navigation policy. Memory helps the robot act when neighbors are not fully observable, while the structured graph assigns different roles to robot history and robot-human interactions. The method avoids requiring an explicit analytic pedestrian transition model or expert initialization as the central training mechanism.

What the evidence establishes. The inspected evaluation varies field of view across 90, 180 and 360 degrees and group populations across 10, 15 and 20 humans, with 500 unseen

cases per setting. A recurrent-attention ablation uses matched PPO settings to better isolate the architecture. Physical TurtleBot 2i demonstrations use RGB-D detection/tracking and a tracking camera, with processing on a remote host.

Critical appraisal. Restricted field of view is not equivalent to arbitrary detector failure, stale tracks or calibrated uncertainty. The limitations explicitly recognize the difference between an invisible simulated robot and reciprocal real interactions. Remote processing also changes the deployment timing boundary. The retrieved preprint is a later version than the conference year; its version must be retained in any detailed comparison.

Implication for novel work. A memory-based comparator prevents the proposed project from treating partial observability as an untouched problem. Test controlled latency and track-age interventions separately from visibility, and distinguish temporal representation gains from low-level response-model gains.

B.3.5 A taxonomy of what prediction models actually assume

Publication: Rudenko et al. [39]. *Human motion trajectory prediction: a survey.* Publication record.

Evidence anchor: Selected full-text inspection. Author survey: taxonomy/classification rules and physics-based, pattern-based and contextual-model discussion. Inspected source copy.

Method and contribution. Rudenko and colleagues organize trajectory prediction around motion models, contextual information and other modeling choices. Physics-based methods encode explicit dynamics; pattern-based methods learn regularities from observed motion; higher-level reasoning can incorporate goals, semantics and interactions. Context is not a single binary attribute: obstacle awareness, map awareness and semantic awareness provide different information and failure modes.

What the evidence establishes. The survey synthesizes methods and datasets across robotics and related communities. Its classification rules explicitly acknowledge mixed methods rather than forcing every approach into one exclusive category. The result is a vocabulary for choosing a forecast appropriate to the task and data, not evidence that the most complex predictor is universally preferable.

Critical appraisal. A forecast is evaluated under a conditioning set. A model supplied with a person's destination cannot be ranked directly against a model using only recent detections. Long-horizon trajectory error, immediate collision coverage and response to a robot intervention are different targets. Survey breadth also means that detailed claims about an individual predictor should be checked against its primary paper.

Implication for novel work. Specify prediction horizon, sensor history, map/goal information and whether the robot's planned action conditions the prediction. Constant velocity should remain an explicit low-cost baseline, with failures analyzed by maneuver and observation quality rather than dismissed categorically.

B.3.6 Social LSTM: recurrent interaction encoding before planning

Publication: Alahi et al. [1]. *Social LSTM: Human Trajectory Prediction in Crowded Spaces.* Publication record.

Evidence anchor: Selected full-text inspection. CVPR paper: problem description, social-pooling formulation, Section 4 and Table 1. Inspected source copy.

Method and contribution. Social LSTM models each pedestrian's history with a recurrent network and pools neighboring hidden states to represent local interaction. It predicts future pedestrian motion from tracked positions, allowing curved and mutually influenced trajectories rather than independent straight-line extrapolation. The social pooling operation is a learned representation of interaction, not a navigation policy or a guarantee that a robot action will be acceptable.

What the evidence establishes. The paper evaluates five ETH/UCY subsets and reports displacement metrics over a 4.8-second forecast. Table 1 notes that the IGP comparator receives the intended destination, unlike other methods. Improvements are not uniform: the average final-displacement entry for Social LSTM is 0.61 versus 0.60 for the listed social-force baseline, despite advantages on other metrics and scenes.

Critical appraisal. The paper illustrates why an abstract-level claim of outperforming prior work needs table-level qualification. Dataset splits, synthetic pretraining, definitions of displacement error and privileged destinations affect comparison. Predicting trajectories from already available tracks does not solve missing detections or identity swaps. A robot also needs to turn the prediction into a feasible decision before its scene changes.

Implication for novel work. Use the method to establish interaction-aware prediction history, not to claim solved closed-loop navigation. Evaluate forecast-to-planner coupling and collision-critical errors, and preserve the conditioning information and metric definition in any reproduced baseline.

B.3.7 Multimodal futures and the limits of best-of-sample evaluation

Publication: Gupta et al. [19]. *Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: model overview, Section 4 protocol, sampling/variety-loss analysis and pooling discussion. Inspected source copy.

Method and contribution. Social GAN uses a recurrent encoder-decoder generator, interaction pooling and an adversarial discriminator to produce several plausible futures from the same observed history. A variety loss encourages at least one sample to match the recorded future. This addresses the inadequacy of averaging mutually exclusive routes into one implausible mean trajectory, while retaining efficient joint prediction across pedestrians.

What the evidence establishes. ETH/UCY evaluation uses leave-one-out scenes, eight observed steps and forecasts of eight or twelve steps at 0.4-second intervals. Quantitative reporting selects the best sampled prediction in an L2 sense. The paper also reports difficulty reproducing the original Social LSTM ranking without its synthetic-data setup, and notes that pooling can improve qualitative coherence without improving every displacement metric.

Critical appraisal. Best-of-N error benefits from increasing sample count and does not reveal which sample a robot should trust before seeing the future. Diversity is not a calibrated probability distribution over dangerous outcomes. A planner that rejects every sampled path can freeze, while selecting the convenient sample can understate risk. The timing cost of generating and checking samples belongs in the closed-loop evaluation.

Implication for novel work. Test whether multimodality improves encounter coverage and collision-free progress under a matched computational budget. Compare against constant velocity and calibrated occupancy regions rather than equating a lower best-of-N score with safer navigation.

B.3.8 Trajectron++: dynamics-aware probabilistic forecasting

Publication: Salzmann et al. [41]. *Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: desiderata and architecture description; Section 5 datasets, metrics and comparator definitions. Inspected source copy.

Method and contribution. Trajectron++ combines a graph-structured recurrent representation with latent-variable forecasting, heterogeneous inputs such as semantic maps, and agent-specific dynamics. Its predictions can be conditioned on a proposed ego trajectory, making the robot's action part of the forecasting problem. This is a more appropriate conceptual interface for interaction-aware planning than assuming that each person's future is independent of what the robot does.

What the evidence establishes. Evaluation spans pedestrian ETH/UCY data and heterogeneous nuScenes traffic. The inspected protocol reports average/final displacement, KDE-based negative log likelihood and best-of-N metrics. Those measures assess different properties of a generative model. The examples demonstrate a flexible forecasting framework, not a closed-loop G1 contact/comfort study.

Critical appraisal. Dynamics-consistent forecasts are not necessarily probability-calibrated under a new domain. Ego-conditioned prediction also depends on the distribution of ego actions represented in the training data: an implausible proposed maneuver can induce an unreliable response forecast. Map availability, object classes and sampling frequency differ considerably from a torso-mounted LiDAR tracker in an online-built map.

Implication for novel work. The paper rules out novelty based simply on combining forecasting with dynamics or ego-plan conditioning. The open question is whether uncertainty and sensing age can be connected to executable humanoid decisions with verified coverage and useful progress.

B.4 Perception and state estimation

B.4.1 Learning range-data detections instead of hand-crafting clusters

Publication: Beyer et al. [4]. *DROW: Real-Time Deep Learning-Based Wheelchair Detection in 2-D Range Data.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: Section II preprocessing/voting and Section III dataset and split design. Inspected source copy.

Method and contribution. DROW builds distance-normalized cutouts around individual laser returns, predicts a class and an offset toward an object's center, and consolidates these votes using a grid and non-maximum suppression. This separates the sensor surface points from the latent object center. It also addresses the changing angular extent of an object with range, which makes a fixed-width raw scan window problematic.

What the evidence establishes. The original paper focuses on wheelchairs and walkers, with approximately 464,000 scans and 24,012 annotated scans from a care facility. Training, validation and test data include spatial separation and temporally disjoint sequences. The described scanner is mounted near the floor, around 37 cm, a key detail for assessing transfer to humanoid sensing.

Critical appraisal. The method is not originally a universal person detector, even though later DROW work extends that scope. Dense scan counts should not be mistaken for equally many independent environments or people. A detector trained on low-mounted 2D geometry may not tolerate torso-height 3D projections, partial upper-body returns or walking-induced sensor motion without adaptation.

Implication for novel work. Compare learned center estimation with geometric clustering using matched projections and ground truth. Measure bias, misses and false positives separately, and verify that perception changes improve downstream navigation rather than only a detector score.

B.4.2 Temporal range features with bounded per-frame work

Publication: Jia et al. [23]. *DR-SPAAM: A Spatial-Attention and Auto-regressive Model for Person Detection in 2D Range Data.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: Section III temporal aggregation; Section IV, Table I and dataset split description. Inspected source copy.

Method and contribution. DR-SPAAM retains a feature template and updates it as new scans arrive. Spatial attention associates nearby features across time, while an autoregressive update accumulates evidence without repeatedly rebuilding cutouts for every previous scan. Its forward-looking aggregation contrasts with backward alignment/recomputation and aims to preserve the benefit of temporal observations at lower computational cost.

What the evidence establishes. The evaluation uses the DROW dataset's person annotations and explicitly separates retrained baselines from previously reported DROW variants. Table I reports AP at a 0.5 m association threshold of 70.3 for DR-SPAAM and 67.9 for the retrained five-scan baseline. Ablations remove spatial attention or autoregression, giving a more informative test than comparing only complete systems.

Critical appraisal. The attribution depends on using the correct retrained baseline and association criterion. Faster network inference is not automatically lower observation-to-command age after ROS transport, queues and planning. A recurrent template can also retain misleading

features; deployment needs a policy for missing scans, reset and stale evidence. The sensor-domain difference from humanoid LiDAR remains important.

Implication for novel work. The architecture offers a concrete lesson for efficient temporal perception. Evaluate throughput, retained-feature age, center/velocity error and navigation outcomes together, using the actual sensor mounting and available compute.

B.4.3 Direct LiDAR-inertial estimation and incremental mapping

Publication: Xu et al. [50]. *FAST-LIO2: Fast Direct LiDAR-Inertial Odometry.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: contribution/estimation discussion and Section VI data-structure benchmark setup. Inspected source copy.

Method and contribution. FAST-LIO2 couples inertial propagation with iterated filtering and direct registration of raw LiDAR returns to a map, avoiding a mandatory hand-engineered feature extractor. An incremental k-d tree supports insertion, deletion, rebalancing and on-tree downsampling. The combination addresses both estimation and the data-structure cost of maintaining useful geometry at high update rates.

What the evidence establishes. The paper separates data-structure evaluation from whole-system odometry comparisons and tests varied LiDAR/IMU datasets and real motions. The inspected setup explains how alternative map structures are integrated and downsampled for comparison. The abstract's fast-update claims therefore belong to identified datasets and hardware, not to an arbitrary ROS graph containing the estimator.

Critical appraisal. This is a state-estimation contribution, not a social navigation controller. Reliable deskew and fusion require acquisition timing, extrinsics and an appropriate IMU contract. A simulator that raycasts a frozen instantaneous scene does not automatically test rolling-scan distortion. Improved trajectory error also does not by itself solve stale person tracks or command-response mismatch.

Implication for novel work. Treat FAST-LIO2 as a controlled estimator alternative. Match sensor sampling and calibration, then compare both localization metrics and changes in navigation decisions. Keep robot-pose truth isolated from onboard control in every experimental arm.

B.4.4 Visual-inertial initialization and persistent map reuse

Publication: Campos et al. [8]. *ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual–Inertial, and Multimap SLAM.* Publication record.

Evidence anchor: Selected full-text inspection. Accepted author manuscript: overview of visual/inertial/multi-map contributions and MAP/IMU-initialization formulation. Inspected source copy.

Method and contribution. ORB-SLAM3 combines feature-based visual estimation, tightly integrated inertial measurements and multiple-map management. The inspected formulation

combines visual reprojection and inertial residuals, with uncertainty-aware maximum-a-posteriori initialization of scale, gravity, velocity and biases. Place recognition and map reuse extend the objective beyond short-term visual odometry.

What the evidence establishes. The manuscript supports several camera configurations and explicitly discusses initialization and data association. Those technical details explain why importing an estimator means more than subscribing to a pose topic. This appraisal does not restate a cross-dataset accuracy multiplier as though it were a measured improvement for Geoff-Sim.

Critical appraisal. Visual observability, calibration, exposure and motion characteristics differ from the project's LiDAR-centric pipeline. Map reuse can help recovery but introduces coordinate corrections that downstream tracks, goals and commands must handle consistently. Dataset trajectory accuracy is not an end-to-end measure of crowd avoidance, and a camera-rich estimator cannot be fairly compared without accounting for its different sensors.

Implication for novel work. Use ORB-SLAM3 to frame alternative sensing and map-lifecycle experiments, not as a presumed drop-in upgrade. The scientific question is which estimation errors actually change collision-critical decisions under matched information and latency budgets.

B.5 Datasets, benchmarks and social validity

B.5.1 Egocentric perception exposes a different failure distribution

Publication: Martín-Martín et al. [32]. *JRDB: A Dataset and Benchmark of Egocentric Robot Visual Perception of Humans in Built Environments.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: dataset overview and detection/tracking results, including Tables 3-4 and occlusion discussion. Inspected source copy.

Method and contribution. JRDB records multimodal observations from a robot's viewpoint and provides human detection/tracking annotations. Panoramic imagery, LiDAR and associated trajectories expose occlusion, moving sensors and dense pedestrian configurations that differ from conventional image collections. Its evaluation decomposes detection quality and tracking consistency instead of assuming that a planner receives an accurate set of human states.

What the evidence establishes. The inspected results use AP and OSPA-style errors for detection, with tracking measures that include false positives, false negatives and identity changes. The discussion shows that methods can rank differently on localization and cardinality error. It also analyzes adaptation to panoramic images, training on JRDB versus generic pretraining, and the sensitivity of results to occlusion.

Critical appraisal. Perception metrics have to be interpreted jointly: a detector that localizes its surviving detections well can still miss many people. Large annotation counts do not eliminate environment or sensor-domain bias. The dataset does not itself establish how a G1's controller responds to those errors, and its sensor arrangement differs from a walking torso-mounted scanner.

Implication for novel work. Borrow the decomposition of missing, misplaced and misassociated people. A navigation evaluation should connect those errors to encounter timing and applied commands, rather than reporting only nearest-track distance or treating detection count as recall.

B.5.2 Demonstration data and a genuine human-evaluation precedent

Publication: Karnan et al. [24]. *Socially CompliAnt Navigation Dataset (SCAND): A Large-Scale Dataset of Demonstrations for Social Navigation.* Publication record.

Evidence anchor: Selected full-text inspection. Author manuscript: dataset description, local-planner evaluation summary/Figure 7, anticipated use cases and conclusion. Inspected source copy.

Method and contribution. SCAND captures human-teleoperated navigation with multimodal sensors and commands on morphologically different robots, including Spot and Jackal. Demonstrations supply behavioral examples when a reward function is hard to specify. The paper also explores demonstrator classification and imitation of global/local planning behavior, recognizing that multiple human strategies can be socially plausible.

What the evidence establishes. The recorded dataset totals 8.7 hours, 138 trajectories and 25 miles. The inspected evaluation reports a study with fourteen human participants for a learned local planner in two scenarios, alongside the demonstration data. It therefore directly contradicts a blanket claim that social-navigation research has no human-subject evaluation.

Critical appraisal. The authors identify geographic and convention bias from collection in Austin, such as passing preferences. Four demonstrators and a finite set of scenarios cannot represent all users or norms. The local-planner study is more relevant to perceived behavior than a pure clearance metric, but it does not establish universal comfort or transfer to a humanoid carrying a payload.

Implication for novel work. Use SCAND as both an imitation-data reference and a methodological comparator for human judgments. The narrower open question is how humanoid morphology, motion transients and task context affect perceived comfort under a controlled navigation comparison.

B.5.3 Grounded scenarios, common metrics and baseline adaptation

Publication: Biswas et al. [5]. *SocNavBench: A Grounded Simulation Testing Framework for Evaluating Social Navigation.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: framework overview, Section 4 algorithms/protocol and Section 5 metric interpretation. Inspected source copy.

Method and contribution. SocNavBench combines curated scenarios grounded in recorded pedestrian data with a shared simulation and evaluation interface. It supports comparing navigation algorithms through several outcome dimensions instead of a single success scalar. Realistic rendering is a capability of the platform, while the experimental modes and information supplied to an algorithm remain separately configurable.

What the evidence establishes. The inspected comparison includes Social Forces, ORCA, SA-CADRL and a pedestrian-unaware planner. Because pedestrians do not react to the robot, the authors adjust ORCA collaboration coefficients. They explicitly discuss kinematic constraints, the use of a meta-planner and selecting only the nearest pedestrians for a fixed-input network. These implementation decisions materially affect what the benchmark tests.

Critical appraisal. A common environment does not automatically mean an equal information or dynamics contract. Recorded pedestrian replay gives repeatability but removes counterfactual reactions to changed robot behavior. Synchronous evaluation can also hide latency effects unless timing is modeled separately. The paper's multidimensional metric analysis is stronger than extracting one aggregate ranking while ignoring these adaptations.

Implication for novel work. Geoff-Sim's benchmark contribution must add a new controlled question, such as delay/response provenance or articulated contact, beyond merely providing worlds and a GUI. Document baseline adapters, actor responsiveness and timing modes with the result.

B.5.4 Configurable multi-robot learning in constrained shared spaces

Publication: Chandra et al. [9]. *SOCIALGYM 2.0: Simulator for Multi-Robot Learning and Navigation in Shared Human Spaces.* Publication record.

Evidence anchor: Selected full-text inspection. AAAI paper: simulator architecture, observations/rewards, environment configuration and limitations. Inspected source copy.

Method and contribution. SocialGym 2.0 emphasizes multiagent learning and configurable indoor interactions such as doorways, corridors, intersections and roundabouts. Observation and reward functions can be customized, along with environmental dimensions, density and robot kinodynamic settings. Its configuration-oriented design makes experimental variation a first-class feature instead of tying every method to one hard-coded social objective.

What the evidence establishes. The short system paper describes extensibility and integration with multiagent learning interfaces. It explicitly notes CPU/parallel-processing limitations and lists continuous actions and further MARL algorithms as future development. Those statements constrain what may be inferred from the platform's advertised flexibility.

Critical appraisal. Configurability is valuable, but each configured benchmark still needs validation. Changing rewards, state observations and dynamics together can undermine algorithm comparisons. Multi-robot interaction is not automatically a model of uncooperative human motion. A simulator architecture paper also supplies a different type of evidence from a statistically characterized physical deployment.

Implication for novel work. Adopt explicit scenario/configuration variation and retain its provenance. Position Geoff-Sim's distinct contribution around measured biped execution and source-audited evaluation, rather than claiming that configurable shared-space benchmarks or mixed dynamics are absent from existing tools.

B.5.5 Compositional urban variation and the success-cost tradeoff

Publication: Wu et al. [49]. *MetaUrban: An Embodied AI Simulation Platform for Urban Micromobility.* Publication record.

Evidence anchor: Selected full-text inspection. arXiv v2: platform overview, Section 4 tasks/protocol and Table 1 benchmarks. Inspected source copy.

Method and contribution. MetaUrban procedurally composes urban spaces, obstacles, moving people and heterogeneous mobility platforms. Its benchmark separates static point navigation from social navigation and compares online RL, safe RL, offline RL and imitation approaches. The inspected main experiment uses a wheeled delivery platform with acceleration, braking and steering actions, rather than a low-level G1 locomotion policy.

What the evidence establishes. Table 1 reports in-distribution and unseen-environment outcomes, including success, path/social metrics and cumulative costs. For the displayed social-navigation test, PPO has higher success than its constrained variants while those variants can reduce cost. The report explicitly discusses the difference between improving safety properties and maintaining effectiveness.

Critical appraisal. The large scene library and asset count do not themselves establish behavioral realism or coverage of rare interaction failures. Generalization to unseen layouts is not identical to generalization to a new controller, sensor mounting or crowd policy. Cost reductions should be interpreted together with mission completion and episode exposure, especially when a method stops or fails early.

Implication for novel work. Use compositional scene variation and held-out layouts, but retain the lower-level G1 execution question. Report a collision-completion frontier rather than treating either safety cost or success alone as the definitive result.

B.5.6 Human-in-the-loop tasks and the meaning of simulated embodiment

Publication: Puig et al. [37]. *Habitat 3.0: A Co-Habitat for Humans, Avatars and Robots.* Publication record.

Evidence anchor: Selected full-text inspection. Author report: Section 3 avatar/simulator design and social-navigation baseline/sensor-ablation discussion. Inspected source copy.

Method and contribution. Habitat 3.0 combines articulated human avatars, efficient appearance/motion generation and interfaces for real people to interact with simulated robots. Social navigation includes finding and following a person, while social rearrangement introduces collaborative task structure. The paper separates skeletal collision representation from skinned visual appearance, which is important when interpreting the physical meaning of a rendered scene.

What the evidence establishes. The inspected navigation experiment uses Spot in train/validation/test scene splits and compares a map-informed heuristic expert with sensor-to-action RL and sensor ablations. The stated RL inputs include depth, a humanoid detector and humanoid GPS. Consequently, the observation contract must be read before calling a method fully unprivileged simply because it also consumes images.

Critical appraisal. Human-in-the-loop simulation can capture responses that prerecorded replay cannot, but it is not identical to in-person interaction with a physical robot. Avatar articulation also does not imply that its gait is a dynamically balanced low-level humanoid controller. Task definitions such as person following differ from independent waypoint avoidance and need distinct success criteria.

Implication for novel work. Habitat provides strong prior art for interactive evaluation and collaborative tasks. Geoff-Sim should claim its specific physics/control and provenance contribution, and use human-in-the-loop tools as a possible evaluation bridge rather than inferring hardware validity from visual realism.

B.5.7 Composed tasks reveal the gap between skills and missions

Publication: Sun et al. [43]. *Beyond Isolation: A Unified Benchmark for General-Purpose Navigation*. Publication record.

Evidence anchor: Selected full-text inspection. arXiv v1: evaluation platform, Tables II-IV and task/metric definitions. Inspected source copy.

Method and contribution. OmniNavBench composes navigation and interaction objectives across different embodiments in an Isaac Sim platform with continuous physical stepping and locomotion controllers. Its tasks include goal-directed navigation, following and question answering, with social constraints applied across tasks. This extends evaluation from an isolated skill to sequential mission execution.

What the evidence establishes. The inspected tables distinguish whole-episode success, subtask completion, continuous progress, following and social intrusion. They use different completion predicates by task; a PointNav subtask's distance tolerance and the final-episode stopping criterion are not interchangeable. The published comparison includes Carter, Aliengo and H1, and shows that partial progress can coexist with very low complete-task success.

Critical appraisal. A personal-space violation ratio is not a count of articulated physical contacts. The platform's reproducibility claims also require reproducing actual initialization, actor schedules and execution, not just retaining a seed. Composite task failures may arise from language, perception, navigation or task transitions; a low total success rate does not identify the bottleneck.

Implication for novel work. The project cannot claim the first physically grounded multi-task navigation benchmark. A meaningful extension would diagnose why complete humanoid missions fail through independent source, timing, control and lifecycle instrumentation, with task-specific validity and stopping criteria.

B.5.8 Dynamic-pedestrian language navigation and metric conflicts

Publication: Dai et al. [14]. *DPed-VLN: A Benchmark for Socially Compliant Vision-and-Language Navigation in Dynamic Pedestrian Environments*. Publication record.

Evidence anchor: Selected full-text inspection. September 2026 arXiv v1: dataset and protocol overview, Tables 2-4 and pedestrian-dynamics ablation. Inspected source copy.

Method and contribution. DPed-VLN couples navigation episodes with ordinary route instructions and prior-augmented language about dynamic pedestrians. ORCA-controlled humanoid actors populate Habitat 3.0 scenes, and the evaluation separates goal completion, efficiency and human-related metrics. The language-conditioned problem is adjacent to, but different from, a LiDAR/Nav2 waypoint controller.

What the evidence establishes. The paper reports 33,093 episodes and paired instruction conditions. Its tables compare imitation and reinforcement learning and show that stronger success does not imply a uniformly lower human-collision rate. The static-versus-dynamic pedestrian ablation exposes a major failure distribution shift even when goal-success changes appear modest.

Critical appraisal. The benchmark is a recent preprint, and the reported baselines/protocol are not independently replicated here. Humanoid pedestrians should not be confused with a low-level humanoid robot policy. Prior-augmented instructions add information that an ordinary goal-only system does not receive. Collision and personal-space metrics need their exact event definitions before comparison with a physics-contact scorer.

Implication for novel work. This is evidence that dynamic human environments and broader task grounding are active research areas, not untouched novelty territory. The useful lesson for Geoff-Sim is to separate source privilege, task success and social-risk metrics and to test dynamic exposure explicitly.

B.6 Locomotion, embodiment and transfer

B.6.1 Rapid adaptation changes the locomotion interface

Publication: Kumar et al. [26]. *RMA: Rapid Motor Adaptation for Legged Robots.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: overview of base policy/adaptation, simulation and hardware evaluation, adaptation analysis and conclusion. Inspected source copy.

Method and contribution. RMA combines a locomotion policy with an adaptation module that infers relevant extrinsic conditions from recent experience. Training can use privileged environment parameters, while deployment must infer their effect from onboard histories. The architecture aims to adapt rapidly to changing payload, friction and terrain rather than requiring a new policy for every condition. This makes locomotion response a state-dependent property rather than a fixed velocity gain.

What the evidence establishes. The paper separates simulated parameter variation from physical payload, step, compliant-surface and outdoor tests. Its simulation table averages three initialized policies and 1,000 episodes per initialization. Hardware trials are much smaller, with some severely failing methods stopped after two attempts to limit damage. Removing adaptation degrades the reported performance.

Critical appraisal. The hardware protocol and simulation denominator cannot be pooled. Staying upright without progressing is not equivalent to successful locomotion, a distinction visible in the ablation discussion. RMA is primarily proprioceptive and does not by itself plan

around people. Adaptation can also change the response model expected by a higher-level planner, so a static calibration may become stale.

Implication for novel work. Evaluate whether a navigation response model must condition on recent policy state, load or terrain. Compare calibrated fixed-policy behavior with adaptation explicitly, rather than treating the low-level policy as an invariant actuator.

B.6.2 Massive parallelism is an experimental enabler, not an acceptance result

Publication: Rudin et al. [40]. *Learning to Walk in Minutes Using Massively Parallel Deep Reinforcement Learning*. Publication record.

Evidence anchor: Selected full-text inspection. Retrieved author preprint v3: training overview, Section 4 parallelism experiments, simulation setup and conclusion. Inspected source copy.

Method and contribution. Rudin and colleagues combine an end-to-end GPU simulation/training pipeline with a terrain curriculum and training choices appropriate to thousands of parallel robots. The work analyzes how environment count, per-environment rollout length and batch size interact. Its contribution is not simply running more copies: insufficient temporal horizon can degrade learning even when total sample count is held fixed.

What the evidence establishes. The study varies parallel robot counts and batch sizes and reports multiple training runs. It identifies a tradeoff region rather than unlimited monotonic benefit from parallelism. The conclusion explicitly states that the aim is fast generation of useful locomotion policies, not the absolute highest robustness achievable with every additional technique.

Critical appraisal. Wall-clock training time is conditional on GPU, simulator, task and selected performance threshold. A quickly trained policy can be a valuable research baseline without establishing rare-failure reliability or social competence. Increasing the number of correlated simulator instances also does not create diverse independent deployment environments by itself.

Implication for novel work. Use efficient training to support controlled ablations, repetitions and explicit initialization-only baselines. For payload work, distinguish gains from architecture, initialization, training duration and integration corrections before attributing success to reinforcement learning.

B.6.3 Belief-state locomotion when exteroception is unreliable

Publication: Miki et al. [33]. *Learning robust perceptive locomotion for quadrupedal robots in the wild*. Publication record.

Evidence anchor: Selected full-text inspection. Author manuscript: recurrent-belief explanation, field deployment results, Figure 4 and discussion/method overview. Inspected source copy.

Method and contribution. Miki and colleagues integrate proprioceptive and exteroceptive observations through a recurrent belief representation. A privileged teacher supports training a

deployable controller that exploits terrain information when it is reliable and falls back toward proprioceptive competence when observations are missing or misleading. This addresses the practical problem that visual geometry and physical support are not equivalent, for example with snow, vegetation or transparent obstacles.

What the evidence establishes. The paper reports varied outdoor and underground deployments and inspects internal belief estimates under sensory failure. Examples include covering sensors, encountering compliant support and revising terrain estimates after contact. These experiments are more informative about sensing robustness than a clean height-map benchmark alone.

Critical appraisal. Contact can legitimately disambiguate whether terrain supports a foot; using contact to resolve uncertainty about a nearby person has a different consequence. The task is terrain locomotion rather than human-centered navigation, and the amount of deployment exposure matters when interpreting zero-fall statements. A recurrent representation also does not automatically expose a calibrated uncertainty estimate to a separate planner.

Implication for novel work. Use selective trust in perception as a design principle, but test the human-avoidance implications separately. The project needs an explicit observation-age and uncertainty interface, not merely a hidden state whose apparent robustness is assumed to transfer.

B.6.4 Planning with a repertoire of executable skills

Publication: Hoeller et al. [22]. *ANYmal parkour: Learning agile navigation for quadrupedal robots*. Publication record.

Evidence anchor: Selected full-text inspection. Corresponding 2023 arXiv manuscript: overview and architecture/skill description; bibliographic citation remains the 2024 journal record. Inspected source copy.

Method and contribution. ANYmal Parkour separates perception, locomotion skills and a high-level navigation policy that selects and commands those skills. Learned scene reconstruction handles noisy or occluded observations, while a repertoire includes walking, jumping, climbing and crouching. The navigation policy reasons in relation to the capabilities available below it rather than selecting a geometric path and hoping a generic controller can realize it.

What the evidence establishes. The inspected manuscript describes training modules in simulation and deploying them on ANYmal D through varied obstacle arrangements. Its platform and behavior repertoire are materially different from a G1 following planar velocity commands. The appraisal concerns the corresponding preprint architecture, not an assumption that every detail matches the final journal version.

Critical appraisal. This is strong prior art against a broad claim that navigation has ignored locomotion capabilities. It does not, however, establish social acceptance or collision-free behavior among unpredictable people. Contact that is intentional and useful during parkour can be unacceptable during social passage, and high-speed terrain success is not a substitute for controlled stopping performance near a person.

Implication for novel work. A response-aware G1 planner should define the executable action repertoire it assumes. Its novelty would need to explain why a calibrated low-dimensional envelope is effective or more auditable for social encounters than established capability-aware hierarchies.

B.6.5 Mobility-aware navigation already studies delay and tracking failures

Publication: Lee et al. [27]. *Learning robust autonomous navigation and locomotion for wheeled-legged robots.* Publication record.

Evidence anchor: Selected full-text inspection. Accepted manuscript: Figure 7 comparison/failure analysis, discussion and hierarchical-learning methods. Inspected source copy.

Method and contribution. Lee and colleagues integrate adaptive wheeled-legged locomotion with a learned local navigation policy and a larger-scale route interface. The hierarchy lets local navigation account for what the locomotion controller can achieve. Memory supports exploration and recovery from local minima; the methods discussion describes staged training and notes that additional alternating fine-tuning gave only marginal benefits in their experiments.

What the evidence establishes. The source reports kilometer-scale missions and a controlled comparison with ten trials per method in a challenging point-goal setup. Figure 7 identifies occlusion, delayed replanning, tracking overshoot and finite-memory failures rather than presenting only successful trajectories. This is directly relevant to claims about planning-execution mismatch.

Critical appraisal. The paper already recognizes delayed and imperfect tracking as navigation bottlenecks, so identifying that phenomenon alone is not novel. Its morphology, mobility modes and route assumptions differ from humanoid crowd navigation. Long missions show valuable system competence but are not equivalent to an exposure-controlled contact-risk comparison across policies.

Implication for novel work. A new contribution must specify a different mechanism or provide a controlled finding about calibration, uncertainty or age, beyond restating delay as a problem. The failure-analysis structure is a good model for linking an observed collision to the actual decision and execution sequence.

B.6.6 ExBody and the upper/lower-body objective split

Publication: Cheng et al. [13]. *Expressive Whole-Body Control for Humanoid Robots.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: retargeting, reference-state initialization, rewards, Table IV ablations and physical examples. Inspected source copy.

Method and contribution. ExBody uses human motion to guide expressive upper-body behavior while relaxing lower-body imitation in favor of robust root-motion objectives. Retargeted motion supplies a structured distribution of commands and initialization states. PPO then balances expression, root movement and regularization instead of forcing the robot to reproduce every human joint trajectory despite morphological mismatch.

What the evidence establishes. The platform is Unitree H1, with a 19-DoF model in the inspected version. Ablations examine reference-state initialization, random goal sampling, full-body tracking and AMP-related rewards. The discussion identifies degenerate strategies such as kneeling while satisfying parts of the objective, showing why survival or episode length alone can be misleading.

Critical appraisal. Expressive movement and social navigation are connected but distinct tasks. Demonstrations of handshakes, gestures or dancing do not quantify perceived comfort during close passage. The observation/action dimensions and gains of H1 are not transferable to a G1 actor merely by copying weights or reward terms. Objective tradeoffs can also change the base response seen by a navigation planner.

Implication for novel work. The paper motivates explicit arm/base coordination and posture-conditioned response tests. It also shows why a payload-policy result needs initialization and command-distribution controls before attributing improvement to a new training objective.

B.6.7 Imitation quality and task success can move in different directions

Publication: Peng et al. [35]. *DeepMimic*. Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: overview, Sections 10-11, Table 4 and initialization/termination ablations. Inspected source copy.

Method and contribution. DeepMimic combines motion-imitation and task objectives to learn physics-based character skills, supported by reference-state initialization and termination rules that discourage invalid strategies. Policies can adapt a reference motion to a task rather than only replaying it. The approach also explores combinations of skills and morphology changes, providing foundational machinery for later humanoid motion learning.

What the evidence establishes. The inspected results are simulated characters, with task-specific comparisons such as striking and throwing. Table 4 shows that the mixture of imitation and task reward changes success differently by task; imitation alone can reproduce appearance while failing the goal. The paper discloses that much of its performance evidence uses one training run because training is expensive.

Critical appraisal. A natural-looking trajectory is not automatically useful, safe or hardware-realizable. Reference-phase synchronization, servo choices, reward weights and training coverage matter. The paper explicitly leaves physical robot deployment as future work in that version, so it should not be cited as direct evidence of G1 sim-to-real transfer.

Implication for novel work. Use it as the foundation for imitation and initialization controls. In social navigation, report task completion, collision/fall outcomes and motion-quality proxies separately; do not let visual resemblance or a longer episode substitute for completing the mission.

B.6.8 Dynamics randomization distributes training assumptions

Publication: Peng et al. [36]. *Sim-to-Real Transfer of Robotic Control with Dynamics Randomization*. Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: Section IV objective, dynamics-parameter distribution and Fetch pushing-task description. Inspected source copy.

Method and contribution. Dynamics randomization optimizes expected return across a distribution of simulator parameters rather than a single nominal model. A policy can use interaction history to adapt implicitly to the latent dynamics of a deployment. The inspected formulation makes the sampling distribution part of the learning problem and motivates recurrence when physical parameters are not directly observable.

What the evidence establishes. The paper demonstrates the approach on a puck-pushing task with a Fetch arm, transferring a simulation-trained policy without physical-system training. This is evidence for a transfer mechanism on that task, not for humanoid walking or human-aware navigation. The physical and observation contracts differ from both RMA and the G1 payload controller.

Critical appraisal. Robustness across sampled parameters does not establish coverage of arbitrary deployment dynamics. Correlations, rare extremes, delays and contact effects outside the training distribution can still matter. Randomizing a parameter also does not supply a calibrated online bound on stopping distance or body occupancy. The distinction between adaptation and certification remains essential.

Implication for novel work. Use randomization as a training factor and preserve its distribution in manifests. Compare it with measured calibration and held-out perturbations, particularly for arm posture, payload inertia and stop/reverse transients that may be poorly represented by nominal locomotion training.

B.6.9 Residual action alignment and downstream policy transfer

Publication: He et al. [20]. *ASAP: Aligning Simulation and Real-World Physics for Learning Agile Humanoid Whole-Body Skills*. Publication record.

Evidence anchor: Selected full-text inspection. arXiv v3: Sections III-IV, Tables III-IV, data collection and sim-to-sim/physical setup. Inspected source copy.

Method and contribution. ASAP first learns motion tracking, collects deployment trajectories, fits a residual action model that makes simulation reproduce observed transitions, and fine-tunes the control policy in that corrected environment. The correction acts through the action interface rather than only changing the next-state prediction. This makes dynamics mismatch experimentally actionable instead of treating the simulator as a fixed ground truth.

What the evidence establishes. The inspected evaluation separates open-loop transition matching from closed-loop motion imitation, compares parameter identification and residual-state alternatives, and uses cross-simulator tests before selected physical G1 tasks. It reports a 43-motion collection with difficulty levels and defines imitation success through body-tracking error, not pedestrian avoidance.

Critical appraisal. Better open-loop matching need not automatically improve closed-loop behavior, which is why the paper's two-level evaluation matters. Its hardware tasks are selected to exercise transfer gaps, not a representative crowd-navigation cohort. A dynamics correction

learned from those motions is not necessarily valid for loaded docking, unseen command reversals or reactive interpersonal encounters.

Implication for novel work. ASAP narrows novelty around learned dynamics correction. Geoff-Sim can instead study a calibrated response envelope for a frozen policy at the navigation interface, but must compare model fidelity and downstream mission outcomes separately and demonstrate value beyond retraining or fixed conservative margins.

B.7 Sensor-level humanoid navigation

B.7.1 Safety-inspired training is not an online invariance certificate

Publication: Wang et al. [48]. *End-to-End Humanoid Robot Safe and Comfortable Locomotion Policy*. Publication record.

Evidence anchor: Selected full-text inspection. arXiv v1: Sections III-V, equations 1-9, Tables I-III and physical deployment description. Inspected source copy.

Method and contribution. Wang and colleagues map temporal LiDAR features and proprioceptive/command history to motor actions. They formulate constrained reinforcement learning with P3O, privileged training critics and barrier-inspired costs. Comfort-oriented terms shape proximity, approach behavior and smoothness. The paper explicitly says that the barrier condition is converted into a learning cost rather than used to filter actions online.

What the evidence establishes. Thirty trials per simulation scenario yield rounded rates including 86% for dynamic agents, 60% for a narrow passage and 83% for a suspended obstacle. The simpler P3O baseline reaches 90% on the suspended-obstacle case. Table II reports nonzero unsafe-region time, with a 0.6 m evaluation threshold distinct from the 0.8 m training distance. G1/Livox deployment is described with onboard computation.

Critical appraisal. The mathematical definition of a safe policy is stronger than what finite-weight approximate learning and these experiments establish. The descriptions of the P3O ablation leave attribution between barrier costs and comfort rewards less clean than a fully factorial design. Some reward-sign conventions merit code-level verification before reuse. Comfort is assessed by proxies rather than a reported participant study in the inspected experiments.

Implication for novel work. This is direct prior art for sensor-driven G1 avoidance. A modular method must compete on matched sensing, timing, contacts and liveness, while stating whether its contribution is interpretability, calibrated uncertainty, compute efficiency or a stronger justified guarantee.

B.7.2 Omni-Perception and the metric ambiguity of success plus collision

Publication: Wang et al. [47]. *Omni-Perception: Omnidirectional Collision Avoidance for Legged Locomotion in Dynamic Environments*. Publication record.

Evidence anchor: Selected full-text inspection. Retrieved arXiv v2: Section 3 architecture; Sections 4-6; Tables 1, 3 and 4; failure cases. Inspected source copy.

Method and contribution. Omni-Perception processes LiDAR history through proximal and distal pathways: near geometry receives more detailed sampling, while farther geometry is compressed. Recurrent features feed a locomotion policy, and a cross-platform raycasting toolkit supports training with sensor-pattern and noise variation. Avoidance rewards include sector-based directional behavior and clearance-related terms alongside commanded motion.

What the evidence establishes. The paper separates LiDAR rendering throughput from policy ablations and physical scenarios. Table 3 uses six dynamic obstacles and thirty trials, reporting 76.7% success and 56.7% collision for the proposed method. Table 4 includes 27/30 successful moving-human trials, but a small-obstacle case favors the native Unitree comparator. The hardware evidence spans legged contexts and must not be assumed to be a matched G1-only benchmark.

Critical appraisal. Success and collision entries in Table 3 are not complementary, and the inspected reporting does not justify converting them into a collision-free completion rate. Similar-looking point clouds are useful qualitative sensor evidence but not a calibrated sensor-error distribution. The limitations identify lost thin-object detail from downsampling and failures in dense vegetation, exposing a perception-compression tradeoff.

Implication for novel work. This paper rules out broad novelty claims about raw-LiDAR end-to-end legged avoidance or multi-simulator LiDAR rendering. A useful project comparison should separate sensor fidelity, runtime age, obstacle recall, goal completion and actual contact, with transparent event definitions.

B.7.3 Generative navigation, dynamic adaptation and cross-embodiment evidence

Publication: Gong et al. [18]. *FLUX: Accelerating Cross-Embodiment Generative Navigation Policies via Rectified Flow and Static-to-Dynamic Learning.* Publication record.

Evidence anchor: Selected full-text inspection. arXiv v1: dynamic-scene protocol, metric definitions, Sections V-VI and Tables II-IV. Inspected source copy.

Method and contribution. FLUX uses rectified-flow trajectory generation and a static-to-dynamic learning curriculum. Visual history and depth condition a transformer-based policy, with dynamic-task refinement after acquiring static geometric priors. DynBench introduces continuing pedestrian activity and structured routes, so the challenge is more than avoiding agents that stop permanently at their destinations.

What the evidence establishes. Table II reports SocialNav success of 64.0 versus 59.3 for NavDP in that protocol. Physical demonstrations include wheeled, quadrupedal and G1 platforms, with RGB-D sensing and remote computation described in the paper. The reported 216.5 ms selected inference configuration uses a different timing boundary and hardware from the project's local planner. These values should not be placed in a common leaderboard.

Critical appraisal. The metric named Collision Count is defined through distance below 0.45 m, not direct contact. The social-distance measure also needs its direction-of-improvement read carefully: time inside a proximity region is not a positive comfort score merely because the metric is named compliance. Short inference improvements do not by themselves quantify acquisition-to-execution freshness or human comfort.

Implication for novel work. FLUX establishes close prior art for cross-embodiment dynamic navigation and G1 transfer. The remaining opening is a precise response/observation contract and auditable downstream evidence, not the first combination of a humanoid and a generative or sensor-driven navigation pipeline.

B.8 Enabling technology and systems

B.8.1 Physics simulation enables experiments but does not validate their model

Publication: Todorov et al. [44]. *MuJoCo: A physics engine for model-based control.* Publication record.

Evidence anchor: Record-limited inspection. Verified 2012 DOI identity; accessible current official overview for software context. Attempts to obtain the original author PDF were unsuccessful.

Method and contribution. MuJoCo is the physics foundation for the project, supporting articulated bodies, contacts, actuation and sensor-related calculations. The current official overview explains its model/data separation and articulated simulation approach. This appraisal distinguishes that accessible software context from the original 2012 paper, whose full experimental text was not recovered during this review.

What the evidence establishes. The primary publication exists and is bibliographically verified. The project can run native simulator tests and compare model behavior, but those checks are evidence about its implementation, not independent validation of every contact or sensor assumption. No numerical claim from the inaccessible original paper is newly attributed here.

Critical appraisal. A physically simulated result remains conditional on geometry, inertial parameters, actuator gains, contact settings, integration and sensor timing. Visual realism, a named physics engine and a high simulation rate answer different questions. Likewise, a magnetic attachment is an explicit development mechanism, not evidence of finger-force closure simply because the rest of the scene obeys physics.

Implication for novel work. Record the actual simulator version and model/patch identity, calibrate critical behavior and separate contact semantics from visual overlays. Simulator integration is enabling engineering; a scientific contribution requires a new question or validated method tested with it.

B.8.2 Middleware timing must be measured at the right boundary

Publication: Macenski et al. [31]. *Robot Operating System 2: Design, architecture, and uses in the wild.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: architecture overview, networking experiments and deployment-case discussion. Inspected source copy.

Method and contribution. ROS 2 supplies modular communication, DDS-based middleware abstraction and lifecycle concepts for composing robotics systems. Its design supports different transport and execution choices rather than prescribing a single timing behavior. The paper

connects those architectural decisions to practical systems and contrasts intra-process, composed and multiprocess communication.

What the evidence establishes. The inspected networking experiments specify hardware, message sizes, publication rates, middleware and packet-loss conditions. Sub-millisecond p95 results in some intra-process conditions are meaningful at that measured boundary. They do not include an arbitrary robot's detector, planner, policy inference, bridge and physical response.

Critical appraisal. Middleware reliability and physical safety are different claims. A successfully delivered message may contain an old observation; a fast callback may wait behind other work; a published target is not a motor acknowledgement. Quality-of-service choices can trade loss, buffering and freshness, so aggregate node timing cannot reconstruct the full causal pipeline without timestamps and identity.

Implication for novel work. Instrument acquisition, decision, command consumption and response separately. A source-audited timing study can be scientifically useful if it tests a new causal hypothesis, while simply using ROS 2 or reporting average callback time remains infrastructure work.

B.8.3 Behavior-tree orchestration and long-run field evidence

Publication: Macenski et al. [30]. *The Marathon 2: A Navigation System.* Publication record.

Evidence anchor: Selected full-text inspection. Author preprint: architecture overview, experimental setup and Tables II-III. Inspected source copy.

Method and contribution. The Marathon 2 presents Navigation2's modular navigation architecture and behavior-tree task orchestration. Planning, control and recovery are composed as interacting services/actions rather than a single monolithic navigation function. That structure makes it possible to adapt the task logic and plugins, but it also creates interfaces whose readiness and timing matter.

What the evidence establishes. The campus experiment uses RB1 and TIAGo platforms, reporting 37.4 combined miles, 22.8 hours and 168 recoveries, with zero recorded collisions and emergency stops in that study. Both robots are limited to 0.45 m/s; one listed configuration uses external navigation computation and another onboard computation. These are important conditions of the field evidence.

Critical appraisal. Long-duration operation is stronger evidence of system integration than a short demo, but it does not establish a humanoid control model or a universal collision probability. Recovery count also matters: successful operation can involve repeated interventions by the autonomy stack without human intervention. Task success should not erase the cost or cause of those recoveries.

Implication for novel work. Treat Nav2 as established infrastructure and a useful source of lifecycle/recovery patterns. Research novelty should concern how measured G1 response, observation freshness or planner-supervisor consistency changes outcomes, not the presence of a behavior tree or repeated waypoint goals.

B.8.4 Mapping modes trade completeness against freshness

Publication: Macenski and Jambrecic [29]. *SLAM Toolbox: SLAM for the dynamic world.* Publication record.

Evidence anchor: Selected full-text inspection. JOSS paper: summary, feature description, graph serialization and synchronous/asynchronous mode discussion. Inspected source copy.

Method and contribution. SLAM Toolbox provides graph-based mapping, localization and persistent map/session operations. The paper highlights serialization of raw data and the pose graph, allowing a session to be resumed or refined. Synchronous operation can queue observations for mapping quality, whereas asynchronous processing favors current localization by admitting new work when the previous update completes.

What the evidence establishes. The inspected feature discussion explains practical mapping and map-maintenance capabilities. It is a software paper, not a randomized controlled comparison of social navigation policies. Its treatment of large and changing environments should therefore be understood as mapping functionality and application experience rather than direct evidence of dynamic-person avoidance.

Critical appraisal. The distinction between map completeness and freshness is central to downstream planning. A good eventual map can coexist with decisions made from old state, while dropping scans can improve freshness but lose geometry. Localization corrections can also move goals and obstacle representations unless the whole pipeline uses consistent frame/time semantics.

Implication for novel work. Record which mapping mode and update policy ran. Test whether map quality, pose corrections or delivered-route age actually explain mission failures instead of assuming that successful SLAM implies reliable navigation through the crowd.

C Claim audit and immediate priorities

Potential claim	Assessment and required evidence
"First humanoid social-navigation system"	Contradicted by direct prior art; do not use.
"First dynamics-aware avoidance"	Contradicted by DWA, VO, biped planning, and subsequent embodied methods.
"Novel response-calibrated uncertainty-aware method"	Possible only after specifying the new mechanism and comparing the nearest methods under matched conditions.
"Sensor-only navigation implemented"	Supported by the documented information boundary and tests, within its stated graph/configuration scope.
"Reliable onboard avoidance achieved"	Not supported by the current documented collisions and timeouts.
"Hard real-time planning"	Not supported by a soft search budget or observed p95 runtime.

Potential claim	Assessment and required evidence
"Formal safety guarantee"	Requires valid model/disturbance assumptions and an appropriate proof; heuristic rollout filtering is insufficient.
"Socially comfortable motion"	Geometric proxies may be reported as proxies; perceived comfort requires human-centered evidence.
"Reproducible benchmark contribution"	Plausible with validated artifacts, controlled replay, external baselines, and new scientific findings; seeded development demos alone are insufficient.
"Simulation results transfer to hardware"	Requires hardware evidence; neither MuJoCo realism nor another paper's G1 deployment establishes this project's transfer.

D Linked bibliography

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